Probability and Statistics

Managing Uncertainty

Year-wise Worked Examination Papers

2022–23 to 2025–26

Questions, detailed solutions, and original papers

An examination study companion

Based on the eight supplied NMIMS examination PDFs.

How to use this volume

Read the question first, cover the solution, and identify the method before calculating. Each worked answer states the model, defines its notation, substitutes the data, and interprets the result. Numerical probabilities are rounded only at the end. Grouped-data calculations use midpoints and are consequently estimates.

All questions are addressed, including the optional questions. Question wording is transcribed in a concise form while preserving the data and tasks; the complete original scans, including their tables and instructions, are reproduced in the appendix. The printed examination instructions describe the original exams; they do not limit the coverage of this study guide.

All hypothesis tests in this edition use t-tests and critical t-values, without p-values. Let α denote the significance level and ν the degrees of freedom. A one-sample test uses $\nu = n - 1$, where n is sample size; a pooled two-sample test is used when equal population variances are assumed, and Welch's two-sample t-test is used otherwise. Normal-distribution probability questions still use standardized normal scores; these are probability calculations rather than hypothesis tests. If an original question does not provide the sample spread needed for a t-test, any extra assumption is stated explicitly.

Source issues. Eight files contain seven distinct papers: the two uploads of the 2025–26 final are byte-for-byte identical. They share one worked chapter; both are retained in the appendix. Papers are ordered by the academic year printed on their cover, then by examination date. A re-exam's eligible batch is also shown. Missing information and inconsistent data are identified explicitly rather than replaced silently.

Diagrams. Statistical class-interval diagrams, plots, and probability diagrams are included where relevant. These papers concern statistics rather than object-oriented classes.

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A focused 70-mark practice set

Study these seven questions first. Each selected question carries 10 marks in its original paper. There is one question from each of the seven distinct papers. Together they form a 70-mark practice set; they do not predict or guarantee 70 marks in a future examination.

The earlier count of 70 repeated marks counted repeat appearances across papers, including two repeats of Bayes' theorem. It was not 70 marks of different material. This selection instead balances the recurring methods, so that practice is not spent repeating the same formula unnecessarily.

Set	Original paper and question	Main skill	Marks
1	2022–23 final, Q3(b)	Normal probability and a percentile	10
2	2022–23 special re-exam, Q3(b)	Grouped mean, spread, mode and skewness	10
3	2023–24 final, Q5(b)	Binomial count; Poisson approximation	10
4	February 2024 re-exam, Q7(b)	Two independent means; critical-value test	10
5	December 2024 final, Q6(a)	Joint, marginal and conditional probabilities	10
6	February 2025 re-exam, Q5(a)	Poisson counts and complements	10
7	2025–26 final, Q2(b)	Bayes' theorem, proof and interpretation	10
		Total	70

How to practise. Attempt each question before reading its solution. In a numerical answer, write the model, define every symbol, substitute the data, and finish with the answer in context. For a test, include hypotheses, assumptions, the critical value and the decision. After checking your work, redo only the steps you could not produce without help.

Coverage limit. This is a starting set, not a replacement for every topic: for example, it does not practise correlation, uniform and exponential distributions, geometric waiting counts or a one-sample t-test. The full year-wise solutions remain after this set for follow-up practice.

Practice 1. Normal probability and an upper cutoff [10 marks]

Source: 2022–23 final, 21 November 2022, Q3(b).

Question. GMAT scores are approximately normally distributed with mean 527 and standard deviation 112. (i) What is the probability that an individual scores above 500? (ii) How high must an individual score to be among the highest 5%?

Step 1: Define the model. Let X be an individual's score, $\mu = 527$ the population mean, and $\sigma = 112$ the population standard deviation. Then $X \sim N(527, 112^2)$; the second entry is the variance, not the standard deviation. Let Z be the standardized score:

$$Z = \frac{X - \mu}{\sigma}.$$

The distribution of Z is normal with mean 0 and standard deviation 1. Write $\Phi(z) = P(Z \leq z)$ for its cumulative probability at value z.

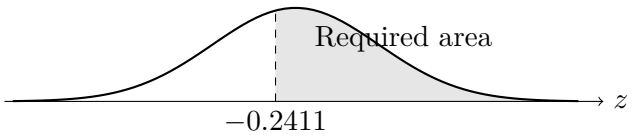
Step 2: Find the probability above 500. Standardize the boundary, not the probability:

$$z_{500} = \frac{500 - 527}{112} = -0.241071.$$

Here z_{500} denotes the standardized value of the score 500. Since the required event is *above* the boundary, take the right-hand area:

$$P(X > 500) = P(Z > -0.241071) = 1 - \Phi(-0.241071) = 0.595250.$$

Thus the probability is about **59.53%**. If a printed table gives only the area between 0 and a positive z-score, use symmetry: the answer is 0.5 plus the area from 0 to 0.241071. Rounding the z-score to 0.24 gives about 0.5948, a normal table-rounding difference.



Step 3: Convert the top 5% into a percentile. Let c be the required cutoff. Only 5% should lie above it, so 95% must lie below it:

$$P(X > c) = 0.05, \qquad P(X \leq c) = 0.95.$$

Let $z_{0.95} = 1.644854$ denote the standard-normal 95th percentile. Reversing the standardization gives

$$c = \mu + \sigma z_{0.95} = 527 + 112(1.644854) = 711.2236.$$

The continuous-model cutoff is **711.22**; if the score must be a whole number, use **at least 712**.

Examination method: For a probability, move from the original score to z. For a percentile, find z first and then move back to the original score.

Practice 2. Karl Pearson skewness [10 marks]

Source: 2022–23 special re-examination, 14 June 2023, Q3(b).

Question. Classes 0–10, ..., 70–80 have frequencies 5, 6, 11, 21, 35, 30, 22, 11. Calculate Karl Pearson skewness.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	5	25	125
15	6	90	1350
25	11	275	6875
35	21	735	25725
45	35	1575	70875
55	30	1650	90750
65	22	1430	92950
75	11	825	61875

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 141$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{6605}{141} = 46.8440.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{350525}{141} - 46.8440^2 = 291.6352.$$

$$\sigma = \sqrt{\sigma^2} = 17.0773.$$

The modal class has lower boundary $L = 40$ and width $h = 10$. Its frequency is $f_m = 35$; the preceding and following frequencies are $f_- = 21$ and $f_+ = 30$. The interpolated mode M_o is

$$M_o = L + h\frac{f_m - f_-}{2f_m - f_- - f_+} = 40 + 10\frac{14}{19} = 47.3684.$$

Karl Pearson's coefficient S_k measures asymmetry in standard-deviation units:

$$S_k = \frac{\bar{x} - M_o}{\sigma} = \frac{46.8440 - 47.3684}{17.0773} = -0.0307.$$

The distribution has negative (left) skewness. A coefficient near zero indicates little asymmetry; its sign describes the longer tail.

How to reproduce the solution. First find the midpoints and the totals in the calculation table. Next compute the mean and standard deviation. Then identify the modal class from the largest frequency and interpolate the mode using its two neighbouring frequencies. Only then substitute into Pearson's coefficient. The result -0.0307 is very close to zero, so the asymmetry is slight; it is not strong left skewness.

Examination method: The modal *class* is 40–50, but the estimated mode is 47.3684. Do not substitute the midpoint 45 in place of the interpolated mode.

Practice 3. Fewer than two defective units [10 marks]

Source: 2023–24 final / 2022–23 re-examination, 25 November 2023, Q5(b).

Question. A company produces units with a defective rate of 3%. In a random sample of 200 independent units, find the probability that fewer than two are defective.

Step 1: Choose the count model. Let X be the number of defective units, $n = 200$ the number inspected, $p = 0.03$ the probability that a unit is defective, and $q = 1 - p = 0.97$ the probability it is satisfactory. Under independent trials with constant defect probability, X has a binomial distribution.

For a possible defective count k,

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

The binomial coefficient counts which k positions are defective, and ! denotes factorial, with $0! = 1$.

Step 2: Translate the words into counts. "Fewer than two" permits only 0 and 1. It excludes 2. These two possible counts cannot occur together, so add their probabilities:

$$P(X < 2) = P(X = 0) + P(X = 1).$$

Substitute each count separately:

$$P(X = 0) = \binom{200}{0}(0.03)^0(0.97)^{200} = (0.97)^{200},$$

$$P(X = 1) = \binom{200}{1}(0.03)(0.97)^{199} = 200(0.03)(0.97)^{199}.$$

Therefore

$$P(X < 2) = (0.97)^{200} + 6(0.97)^{199} = 0.0162483.$$

The required probability is about **1.625%**.

Step 3: Recognize the Poisson approximation. Because n is large and p small, the binomial count can be approximated by a Poisson count with mean $\lambda = np = 200(0.03) = 6$. Here λ is the expected number of defectives. Using the approximation,

$$P(X < 2) \approx e^{-\lambda}\frac{\lambda^0}{0!} + e^{-\lambda}\frac{\lambda^1}{1!} = e^{-6}(1 + 6) = 0.0173513.$$

Here e is the base of natural logarithms. The approximation gives about 1.735%; it is close but not identical to the exact binomial result. Since the question does not require an approximation, report the binomial result as the main answer.

Examination method: Write the allowed counts before choosing calculator functions. The mean count is 6, so fewer than two defectives should be an unusually low count, consistent with the small probability.

Practice 4. Testing the difference between two means [10 marks]

Source: 2023–24 re-examination for 2022–23, 8 February 2024, Q7(b).

Question. Two independent samples have the following summaries. Sample 1: size 60, mean 86, sample standard deviation 6. Sample 2: size 75, mean 82, sample standard deviation 9. (a) Estimate the standard error of the difference between means. (b) At significance level 0.01, test whether the population means are equal.

Step 1: State notation, hypotheses and assumptions. Let $n_1 = 60, n_2 = 75$ be the sample sizes, $\bar{x}_1 = 86, \bar{x}_2 = 82$ the sample means, $s_1 = 6, s_2 = 9$ the sample standard deviations, and μ_1, μ_2 the population means. The samples must be independent, with observations sampled independently within each group. Both samples are large, allowing a normal approximation for their means.

The word "same" requires a two-sided alternative:

$$H_0 : \mu_1 - \mu_2 = 0, \qquad H_1 : \mu_1 - \mu_2 \neq 0.$$

Here H_0 is the null hypothesis, H_1 the alternative, and $\alpha = 0.01$ is the significance level.

Step 2: Compute the standard error. Let SE denote the estimated standard deviation of the sampling distribution of the mean difference. Independence allows the two variance contributions to be added:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{36}{60} + \frac{81}{75}} = \sqrt{0.60 + 1.08} = 1.296148.$$

This answers part (a). Do not subtract the standard deviations or divide their sum by the sample size.

Step 3: Calculate the two-sample t statistic. Population standard deviations are unknown and equality of population variances is not stated. Use Welch's t-test, which uses a separate sample variance for each group. The hypothesized difference under H_0 is zero:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} = \frac{86 - 82}{1.296148} = 3.086067.$$

Step 4: Find the degrees of freedom and critical values. Let $v_1 = s_1^2/n_1 = 0.60$ and $v_2 = s_2^2/n_2 = 1.08$ be the estimated variance contributions. Welch's approximate degrees of freedom ν are

$$\nu = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)} = \frac{1.68^2}{0.60^2/59 + 1.08^2/74} = 129.08976.$$

For a two-sided test at $\alpha = 0.01$, each tail contains $\alpha/2 = 0.005$. The critical values are $\pm t_{0.995,\nu} = \pm 2.61445$, where $t_{0.995,\nu}$ denotes the 99.5th percentile of a t-distribution with ν degrees of freedom. With an integer-degree table, use 129; if that row is absent, round down to the next available row for a conservative comparison.

Step 5: Compare and conclude. Reject H_0 when $|t| > 2.61445$. Here $3.086067 > 2.61445$, so **reject equality of the population means at the 1% level**. The data support a difference, with the first sample having the higher observed mean. This test compares means, not whether the populations are identical in every respect.

Examination method: Write the tail direction, degrees of freedom and critical t-value before making the decision. Large samples do not require changing to a different testing method in this guide.

Practice 5. Joint, marginal and conditional probabilities [10 marks]

Source: 2024–25 final / earlier-batch re-examination, 11 December 2024, Q6(a).

Question. $p(x, y) = k(2x+y)$ for x, y=0, 1, 2. Find k, marginals, independence, and conditional PMF of Y given X=x.

Worked solution. Although the paper calls this a density, the listed support is discrete, so it is a joint PMF. Let U, V denote the two variables (called X, Y or A, B in the question), and let $u, v \in \{0, 1, 2\}$. The constant k normalizes $p(u, v) = k(2u + v)$.

$$1 = k \sum_{u=0}^{2} \sum_{v=0}^{2} (2u + v) = 27k, \qquad k = 1/27.$$

The joint masses and row totals are:

$u \backslash v$	0	1	2	$p_U(u)$
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
$p_V(v)$	6/27	9/27	12/27	1

Marginalization means adding over the other variable:

$$p_U(u) = \sum_{v=0}^{2} p(u, v) = \frac{6u + 3}{27}, \qquad p_V(v) = \sum_{u=0}^{2} p(u, v) = \frac{6 + 3v}{27}.$$

Here p_U, p_V are the marginal PMFs. Since $p(0, 0) = 0$ but $p_U(0)p_V(0) = 2/81 > 0$, the variables are not independent. The conditional PMF of V given $U = u$ is

$$p_{V|U}(v \mid u) = \frac{p(u, v)}{p_U(u)} = \frac{2u + v}{6u + 3}, \qquad v = 0, 1, 2.$$

For every allowed u, these three probabilities sum to one.

Why each operation is different. Normalization adds all nine cells to establish the constant. A marginal distribution adds out the variable you do not want. A conditional distribution keeps the specified row and divides its entries by that row's total. Independence asks whether every joint mass equals the product of its two marginals; a single counterexample is enough to disprove it.

Examination method: After calculating a conditional row, add its probabilities. The result must be 1. Discrete probabilities are summed, not integrated.

Practice 6. Poisson defects and complements [10 marks]

Source: 2024–25 re-examination, 22 February 2025, Q5(a).

Question. Two percent of test tubes are defective. Use a Poisson distribution to find the probability that a packet of 100 contains (i) no defectives, (ii) exactly one, (iii) two or more, and (iv) two or fewer defectives.

Step 1: Find the mean count. Let X be the number of defective tubes, $n = 100$ the tube count, and $p = 0.02$ the defective probability. The Poisson approximation uses expected defective count $\lambda = np = 2$. For a nonnegative integer k,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-2}\frac{2^k}{k!}.$$

Here $g(k)$ is the probability of count k, e the natural exponential base, and $k!$ its factorial. Independent equal-probability defects give the underlying binomial model; the question explicitly asks for its Poisson approximation.

Step 2: Calculate the small counts.

$$g(0) = e^{-2} = 0.135335, \quad g(1) = 2e^{-2} = 0.270671, \quad g(2) = \frac{2^2}{2!}e^{-2} = 0.270671.$$

Therefore the answers to (i) and (ii) are **0.135335** and **0.270671** respectively.

Step 3: Use the correct endpoint and complement. "Two or more" means $X \geq 2$. Its complement is $X \leq 1$, so

$$P(X \geq 2) = 1 - [g(0) + g(1)] = 1 - 3e^{-2} = \boxed{0.593994}.$$

"Two or fewer" means $X \leq 2$, so include the mass at 2:

$$P(X \leq 2) = g(0) + g(1) + g(2) = 5e^{-2} = \boxed{0.676676}.$$

The last two answers do not add to 1 because both events include $X = 2$. They overlap rather than forming complementary events.

Examination method: For "at least r", subtract counts 0 through $r - 1$ from 1. For "at most r", add counts 0 through r. Here r is the stated count threshold.

Practice 7. Bayes' theorem and proof [10 marks]

Source: 2025–26 final, 29 November 2025, Q2(b).

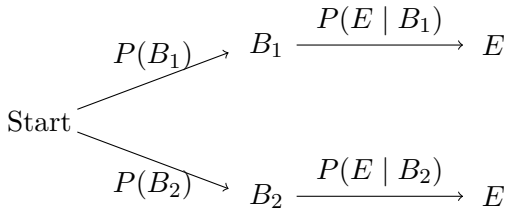
Question. State Bayes theorem; explain its components, proof and applications.

Worked solution. Let $B_1, \ldots, B_m$ be mutually exclusive and exhaustive possible sources, each with positive prior probability. Let E be observed evidence with $P(E) > 0$. Bayes' theorem states

$$P(B_j \mid E) = \frac{P(B_j)P(E \mid B_j)}{\sum_{i=1}^{m} P(B_i)P(E \mid B_i)}.$$

Here m is the number of sources, i indexes all sources, j identifies the source of interest, $P(B_j)$ is its prior probability, $P(E \mid B_j)$ is the likelihood of the evidence, and $P(B_j \mid E)$ is the revised or posterior probability. The denominator is $P(E)$.

For the proof, the definition of conditional probability gives $P(B_j \mid E) = P(B_j \cap E)/P(E)$. The multiplication rule gives $P(B_j \cap E) = P(B_j)P(E \mid B_j)$. Because the sources partition the sample space, the events $E \cap B_i$ are disjoint and their probabilities sum to $P(E)$. Substitution gives the displayed result. It is used to identify a likely source after observing an outcome, for example a defective product's factory or the box from which a coloured ball was drawn.A probability tree organizes the calculation: multiply along a branch to get a joint probability, add evidence branches to obtain the denominator, and divide the selected branch by that denominator.



Only the evidence branches are shown. Each source also has a complementary branch for E^c, the absence of the evidence.

A numerical application to remember. Suppose three equally likely boxes have red-ball fractions $0.5, 0.2, 0.7$. With B_i denoting box i and R a red draw, the probability of box 3 after seeing red is

$$P(B_3 \mid R) = \frac{(1/3)(0.7)}{(1/3)(0.5) + (1/3)(0.2) + (1/3)(0.7)} = \frac{1}{2}.$$

This illustrates the theorem; it is not an additional question counted in the 70 marks. The numerator is the joint probability of the selected source and evidence. The denominator includes every source that could have produced the evidence.

Examination method: For a theory answer, include the assumptions that the source events form a partition and that the observed event has positive probability. Then give the formula, define its components, prove it, and explain an application.

1 2022–23: Final examination, 21 November 2022

Source P1. Academic year 2022–23; all seven questions are solved below.

Q1(a). Bursting pressure of bags

Question. Pressure classes: 5–10, 10–15, 15–20, 20–25, 25–30, 30–35. Frequencies for A: 2, 9, 29, 54, 11, 5; for B: 9, 11, 18, 32, 27, 13. Which has the higher average and greater uniformity? Which would a buyer prefer at equal prices?

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
7.5	2	15	112.5
12.5	9	112.5	1406.25
17.5	29	507.5	8881.25
22.5	54	1215	27337.5
27.5	11	302.5	8318.75
32.5	5	162.5	5281.25

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 110$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{2315}{110} = 21.0455.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{51337.5}{110} - 21.0455^2 = 23.7934.$$

$$\sigma = \sqrt{\sigma^2} = 4.8778.$$

For manufacturer B, apply the same calculation:Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
7.5	9	67.5	506.25
12.5	11	137.5	1718.75
17.5	18	315	5512.5
22.5	32	720	16200
27.5	27	742.5	20418.8
32.5	13	422.5	13731.2

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 110$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{2405}{110} = 21.8636.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{58087.5}{110} - 21.8636^2 = 50.0496.$$

$$\sigma = \sqrt{\sigma^2} = 7.0746.$$

Relative consistency is measured by $CV = 100\sigma/\bar{x}$, where CV is the coefficient of variation in percent, σ is the standard deviation and $\bar{x}$ the mean pressure. A has $CV \approx 23.18\%$ and B has $CV \approx 32.36\%$ (use the unrounded calculations above). B has the slightly higher mean; A is more uniform. There is no unique preference without stating the buyer's objective: choose B for average strength alone, or A if consistency and avoiding low-strength bags are the priority.

Q1(b). Successive dice rolls

Question. Two fair dice are rolled on each independent roll. Find the probability of (i) sum 7 then sum 11; (ii) total 21 across two rolls; (iii) total 6 across three rolls.

Worked solution. Each ordered die pair has probability 1/36. There are six ways to obtain 7 and two ways to obtain 11. Independence therefore gives $(6/36)(2/36) = 1/108 = 0.009259$ for (i).

For (ii), four independent dice must sum to 21. Replace each die score d by $7-d$; their transformed scores must sum to 7. After subtracting 1 from each, four nonnegative integers sum to 3. There are $\binom{6}{3} = 20$ arrangements, none exceeding the die limit. Thus the probability is $20/6^4 = 5/324 = 0.015432$. Here d is an individual die score and the binomial coefficient counts the arrangements.

For (iii), six dice have minimum possible total 6. Every die must therefore show 1, giving $(1/6)^6 = 1/46656 = 0.0000214335$.

Q1(c). Frequency polygon

Question. Briefly describe frequency polygons.

Worked solution. A frequency polygon joins points whose horizontal coordinates are class midpoints and whose vertical coordinates are class frequencies. To close it, add a zero-frequency midpoint one class width before the first class and one after the last. Unlike a bar chart, its horizontal axis represents a numerical scale. It shows the shape and concentration of grouped data and can compare distributions on the same axes. See the wage polygon in the 2025–26 chapter for a complete example.

Q1(d). Number of cars

Question. Couple responses for number of cars are 0, 0.5, 1, 1.5, 2, 2.5 with frequencies 2, 14, 23, 7, 4, 2. Calculate variance and standard deviation.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
0	2	0	0
0.5	14	7	3.5
1	23	23	23
1.5	7	10.5	15.75
2	4	8	16
2.5	2	5	12.5

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 52$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{53.5}{52} = 1.0288.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{70.75}{52} - 1.0288^2 = 0.3021.$$

$$\sigma = \sqrt{\sigma^2} = 0.5496.$$

Here the listed values are exact responses rather than class midpoints, so no grouping approximation is needed. The variance uses the distribution divisor N. If estimating a population variance from a sample, multiply this variance by $52/51$.

Q2(a). Press lifetime: known population spread

Question. Claimed mean lifetime 14500 hours; known population standard deviation 2100 hours. A sample of 25 presses has mean 13000 hours. At $\alpha = 0.01$, is the mean below 14500?

Worked solution. Data assumption for a t-only solution. The original question labels 2100 hours as the *population* standard deviation, and does not give a sample standard deviation. An exact one-sample t-test cannot be recovered from that information alone. To practise the syllabus method, additionally assume that the sample standard deviation is $s = 2100$ hours. The calculation and conclusion below are conditional on that added assumption; the original question's data remain unchanged.

Let μ be the population mean lifetime, $\mu_0 = 14500$ the claimed mean, $\bar{x} = 13000$ the sample mean, $n = 25$ the sample size, and $s = 2100$ the assumed sample standard deviation. Assume independent observations from an approximately normal population. Test $H_0 : \mu = 14500$ against $H_1 : \mu < 14500$ at significance level $\alpha = 0.01$.

$$SE = \frac{s}{\sqrt{n}} = \frac{2100}{5} = 420, \qquad t = \frac{\bar{x} - \mu_0}{SE} = \frac{13000 - 14500}{420} = -3.57143.$$

Here SE is the estimated standard error. The degrees of freedom are $\nu = n-1 = 24$. The lower-tail 1% critical value is $-t_{0.99,24} = -2.49216$, where $t_{0.99,24}$ is the 99th percentile of the t-distribution with 24 degrees of freedom. Since $-3.57143 < -2.49216$, reject H_0. Under the stated sample-SD assumption, the data support mean lifetime below 14500 hours. Without a sample SD or the individual observations, the original data do not specify this t-test completely.

Q2(b). Correcting an erroneous observation

Question. For 100 persons, the reported mean is 32.02 and standard deviation is 13.18. An age of 57 was misread as 27. Find the corrected mean and standard deviation.

Worked solution. Let $N = 100$ be the count, S the sum of ages and Q the sum of squared ages; subscripts o, c mean original and corrected. With the descriptive variance convention,

$$S_o = N\bar{x}_o = 100(32.02) = 3202, \quad Q_o = N(\sigma_o^2 + \bar{x}_o^2) = 119903.28.$$

Here $\bar{x}_o = 32.02$ is the original mean and $\sigma_o = 13.18$ the original standard deviation. Replace the erroneous contribution in both sums:

$$S_c = 3202 - 27 + 57 = 3232, \qquad Q_c = 119903.28 - 27^2 + 57^2 = 122423.28.$$

Thus the corrected mean $\bar{x}_c$ and variance σ_c^2 are

$$\bar{x}_c = S_c/N = 32.32, \qquad \sigma_c^2 = Q_c/N - \bar{x}_c^2 = 179.6504.$$

The corrected standard deviation is $\sigma_c = \sqrt{179.6504} = 13.4034$ years. The calculation treats the printed summaries as raw-data summaries; if they were only midpoint estimates, an exact individual-age correction cannot be recovered without further class information.

Q3(a). Cumulative frequencies and variation

Question. Less-than cumulative counts up to 10, 20, 30, 40, 50, 60, 70, 80 marks are 12, 30, 65, 107, 157, 202, 222, 230. Find standard deviation and coefficient of variation.

Worked solution. Take successive differences to obtain class frequencies 12,18,35,42,50,45,20,8. Assume the first class is 0–10.Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	12	60	300
15	18	270	4050
25	35	875	21875
35	42	1470	51450
45	50	2250	101250
55	45	2475	136125
65	20	1300	84500
75	8	600	45000

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 230$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{9300}{230} = 40.4348.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{444550}{230} - 40.4348^2 = 297.8544.$$

$$\sigma = \sqrt{\sigma^2} = 17.2585.$$

The coefficient of variation is $CV = 100\sigma/\bar{x}$, where CV measures relative spread. Substituting the above values gives approximately 42.68%.

Q3(b). GMAT normal model

Question. GMAT scores are approximately normal with mean 527 and standard deviation 112. Find the probability of a score above 500 and the minimum score for the highest 5%.

Worked solution. Let X denote the measured quantity, with population mean $\mu = 527$ and standard deviation $\sigma = 112$. Standardize using

$$Z = \frac{X - \mu}{\sigma}.$$

Here Z has a standard normal distribution, and $\Phi(z) = P(Z \leq z)$ is its cumulative probability. Convert each endpoint separately, then subtract the cumulative probabilities.

$$P(X > 500) = 1 - \Phi(-0.2411) = 0.595250.$$

Let c be the upper-5% cutoff and $z_{0.95} = 1.64485$ the 95th standard-normal percentile. Then $c = \mu + z_{0.95}\sigma = 527 + 1.64485(112) = 711.2236$. For whole-number scores, a score of at least 712 exceeds the continuous-model cutoff.

Q4(a). Sales training: one-sample t-test

Question. Before training, mean sale was \$100. For 25 salespeople after training, mean sale is \$130 and sample standard deviation \$15. Test whether mean sales increased at 5%.

Worked solution. Let μ be the post-training population mean, $\mu_0 = 100$ the benchmark, $\bar{x} = 130$ the sample mean, $s = 15$ the sample standard deviation and $n = 25$. Test $H_0 : \mu = 100$ against $H_1 : \mu > 100$. Independent approximately normal observations are assumed.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{130 - 100}{15/5} = 10.$$

The degrees of freedom are $\nu = n - 1 = 24$. The upper-tail 5% critical value is $t_{0.95,24} = 1.71088$. Since $10 > 1.71088$, reject H_0: mean sales are higher than the historical benchmark. This comparison alone does not establish that training caused the increase.

Q4(b). Poisson job arrivals

Question. Jobs arrive at mean 2 per minute. Find probabilities of zero jobs, exactly two, at most three, more than three, and a 90% upper count bound.

Worked solution. Let X be the event count in the stated interval, $\lambda = 2$ its mean count, k a nonnegative integer, and e the base of natural logarithms. With independent arrivals at a constant average rate,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-2}\frac{2^k}{k!}.$$

$$P(X = 0) = g(0) = 0.135335.$$

$$P(X = 2) = g(2) = 0.270671.$$

$$P(X \le 3) = \sum_{k=0}^{3} g(k) = 0.857123.$$

$$P(X > 3) = 1 - \sum_{k=0}^{3} g(k) = 0.142877.$$

For a 90% upper count bound, let m be the smallest integer with $P(X \le m) \ge 0.90$. Since $P(X \le 3) = 0.857123$ and $P(X \le 4) = 0.947347$, the bound is **4 jobs**. This is a 90% coverage bound, not an absolute maximum: a Poisson count has no finite maximum.

Q5(a). Conditional dice probabilities

Question. For two dice let A be sum 3, B sum 7, and C at least one die showing 1. Find P(A given C), P(B given C), and test independence of A, C and B, C.

Worked solution. The condition C leaves 11 equally likely ordered outcomes: six with the first die 1, six with the second die 1, counting $(1, 1)$ only once. Both outcomes for A, namely $(1, 2), (2, 1)$, lie in C. For B, only $(1, 6), (6, 1)$ lie in C.

$$P(A \mid C) = \frac{P(A \cap C)}{P(C)} = \frac{2/36}{11/36} = \frac{2}{11}, \quad P(B \mid C) = \frac{2/36}{11/36} = \frac{2}{11}.$$

Here $P(D \mid C)$ means probability of event D after restricting the sample space to C, and $\cap$ denotes intersection. Independence requires $P(D \mid C) = P(D)$. Since $P(A) = 2/36$ and $P(B) = 6/36$, neither pair is independent.

Q5(b). Football attendance

Question. Attendance has mean 16050 and standard deviation 2500. Find probabilities of above 20000, below 10000, and between 14000 and 17500.

Worked solution. Assume a normal attendance distribution; the two summary measures alone do not determine these probabilities.Let X denote the measured quantity, with population mean $\mu = 16050$ and standard deviation $\sigma = 2500$. Standardize using

$$Z = \frac{X - \mu}{\sigma}.$$

Here Z has a standard normal distribution, and $\Phi(z) = P(Z \leq z)$ is its cumulative probability. Convert each endpoint separately, then subtract the cumulative probabilities.

$$P(X > 20000) = 1 - \Phi(1.5800) = 0.057053.$$

$$P(X < 10000) = \Phi(-2.4200) = 0.007760.$$

$$P(14000 < X < 17500) = \Phi(0.5800) - \Phi(-0.8200) = 0.512935.$$

Q6(a). Allergy test and Bayes rule

Question. Allergy prevalence is 1%. Sensitivity is 80%; the false-positive rate is 10%. If a test is positive, what is the probability of allergy?

Worked solution. Let A mean allergy and $+$ a positive test. The joint probability of allergy and a positive result is $P(A)P(+ \mid A) = 0.01(0.8) = 0.008$. The joint probability of no allergy and a positive result is $P(A^c)P(+ \mid A^c) = 0.99(0.1) = 0.099$, where A^c is the complement of A.

$$P(A \mid +) = \frac{0.008}{0.008 + 0.099} = \frac{8}{107} = 0.074766.$$

Thus about 7.48% of positive results correspond to an allergy under this model. Low prevalence means false positives can outnumber true positives.

Q6(b). Rain after a cloudy morning

Question. 50% of rainy days begin cloudy, 40% of all days begin cloudy, and 10% of days are rainy. Find the chance of rain given cloud.

Worked solution. Let R mean rain and C cloudy morning. Given $P(C \mid R) = 0.5$, $P(C) = 0.4$ and $P(R) = 0.1$, Bayes' rule gives

$$P(R \mid C) = \frac{P(C \mid R)P(R)}{P(C)} = \frac{0.5(0.1)}{0.4} = 0.125.$$

The chance is 12.5%. Do not reverse the conditional probability directly: $P(C \mid R)$ and $P(R \mid C)$ answer different questions.

Q7(a). Continuous and joint distributions

Question. Explain continuous random variables, their PDF, and the joint distribution of two continuous random variables.

Worked solution. For a discrete random variable X, the probability mass function is $p_X(x) = P(X = x)$. Here x is a possible value; every mass is nonnegative and the masses sum to one. For a continuous X, the probability density function $f_X(x)$ is nonnegative and has total integral one. Probabilities are areas, not individual density heights:

$$P(a < X \leq b) = \int_a^b f_X(u)\,du.$$

Here a, b are interval endpoints and u is the integration variable. A single point has probability zero in the continuous case. The cumulative distribution function is $F_X(x) = P(X \leq x)$, so

$$F_X(x) = \sum_{u \leq x} p_X(u) \quad \text{(discrete)}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(u)\,du \quad \text{(continuous)}.$$

Where the derivative exists, $f_X(x) = F_X'(x)$. For example, a fair die has mass $1/6$ at each integer from 1 to 6; a uniform measurement on $[0, 1]$ has density 1 inside that interval and CDF equal to 0 below 0, x between 0 and 1, and 1 above 1.For two continuous variables X, Y, a joint density $f_{X,Y}(x, y)$ is nonnegative and integrates to one over the plane. Here x, y are possible values. For a region D,

$$P((X, Y) \in D) = \iint_D f_{X,Y}(x, y)\,dx\,dy.$$

The marginal densities are $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,dy$ and $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\,dx$. The joint CDF is $F_{X,Y}(a, b) = P(X \leq a, Y \leq b)$, obtained by integrating the joint density over $x \leq a, y \leq b$, where a, b are upper limits. Independence means $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ almost everywhere.

Q7(b). Bounced checks

Question. Each of 200 checks independently bounces with probability 0.03. Find probabilities of exactly 10 and exactly 5 bounced checks.

Worked solution. Let X be the number of successes in $n = 200$ independent trials, each with success probability $p = 0.03$. The failure probability is $1 - p = 0.97$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 10) = b(10) = 0.040652.$$

$$P(X = 5) = b(5) = 0.162250.$$

The binomial results are exact under independence. The Poisson approximation uses $\lambda = np = 6$ and gives respectively $e^{-6}6^{10}/10! \approx 0.041303$ and $e^{-6}6^5/5! \approx 0.160623$; λ is the expected bounce count. Label these as approximations if using them in an exam.

2 2022–23: Special re-examination, 14 June 2023

Source P7. The final two income subparts and the machine question contain omissions, addressed below.

Q1(a). Missing accident frequencies

Question. Accident counts 0, 1, 2, 3, 4, 5 have frequencies 46, a, b, 25, 10, 5. There are 200 days and the mean is 1.46. Find the missing counts.

Worked solution. Let a, b be frequencies for one and two accidents; $N = 200$ is the total number of days. The total-frequency equation is $a+b = 200-(46+25+10+5) = 114$. Since the weighted accident total is $N\bar{x} = 200(1.46) = 292$, where $\bar{x}$ is the mean,

$$a + 2b + 3(25) + 4(10) + 5(5) = 292, \qquad a + 2b = 152.$$

Subtracting the first equation gives $b = 38$ and then $a = 76$. Both are nonnegative integers and reproduce the stated total and mean.

Q1(b). Discount brokers

Question. 30% of investors use discount brokers. Among 9 independent investors find probabilities of exactly 2, at most 3, and at least 3 users.

Worked solution. Let X be the number of successes in $n = 9$ independent trials, each with success probability $p = 0.3$. The failure probability is $1 - p = 0.7$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 2) = b(2) = 0.266828.$$

$$P(X \le 3) = \sum_{k=0}^{3} b(k) = 0.729659.$$

$$P(X \ge 3) = \sum_{k=3}^{9} b(k) = 0.537169.$$

Q1(c). A randomly selected machine

Question. Of five machines, three are working properly (item imperfection probability 0.1), two defective (probability 0.2). One machine is chosen randomly and five of its articles examined.

Worked solution. **Source omission:** the printed question stops after "examined" and asks for no event. A unique requested numerical answer cannot be inferred. The following model solves any specified event without inventing a missing condition.

Let G mean a good machine, D a defective machine, and K the number of imperfect articles among the five. Given the machine, article outcomes are independent. For $k = 0, 1, \ldots, 5$,

$$P(K = k) = \frac{3}{5}\binom{5}{k}0.1^k 0.9^{5-k} + \frac{2}{5}\binom{5}{k}0.2^k 0.8^{5-k}.$$

Here $P(G) = 3/5$ and $P(D) = 2/5$ are the selection probabilities. This is a mixture of two binomial distributions; the five outcomes are not independent after the common machine identity is hidden. If k imperfect articles are observed, Bayes' rule gives

$$P(D \mid K = k) = \frac{(2/5)0.2^k 0.8^{5-k}}{(3/5)0.1^k 0.9^{5-k} + (2/5)0.2^k 0.8^{5-k}}.$$

For illustration only, $P(K = 0) = 0.485366$ and $P(D \mid K = 0) = 0.27005$ approximately. These examples are not a replacement for the absent question.

Q1(d). Typist consistency

Question. Typist A: mean 30 pages, SD 6. Typist B: mean 45 pages, SD 10. Which is more consistent?

Worked solution. Use relative rather than absolute spread because the mean outputs differ. Let $CV = 100s/\bar{x}$, where s is standard deviation and $\bar{x}$ mean daily output.

$$CV_A = 100(6/30) = 20\%, \qquad CV_B = 100(10/45) = 22.2222\%.$$

A has the smaller coefficient of variation and therefore greater relative consistency, although B has the higher mean output.

Q2(a). Missing frequencies from median and mode

Question. Classes 10–20, ..., 80–90 have frequencies 3, 7, a, b, c, 20, 16, 4. Total 150, median 48.25 and mode 44. Find a, b, c.

Worked solution. Let a, b, c denote the missing frequencies of classes 30–40, 40–50 and 50–60. The total gives $a + b + c = 100$. The stated median places the median class at 40–50, with lower limit $L = 40$, width $h = 10$, total $N = 150$ and preceding count $10 + a$. Grouped median interpolation gives

$$48.25 = 40 + 10\frac{75 - 10 - a}{b}, \qquad a + 0.825b = 65.$$

The modal class is also 40–50, so the mode formula gives

$$44 = 40 + 10\frac{b - a}{2b - a - c}, \qquad b - 3a + 2c = 0.$$

The three linear equations yield $a = 32, b = 40, c = 28$. Check: $40 + 10(75 - 42)/40 = 48.25$ and $40 + 10(40 - 32)/(80 - 32 - 28) = 44$. In these formulas the median and mode are interpolated positions, not exact individual data values.

Q2(b). An average for open-ended classes

Question. Item sizes below 50, 50–100, 100–150, 150–200, 200 and above have frequencies 15, 20, 36, 40, 10. Find the most suitable average.

Worked solution. The median is suitable because it does not require midpoints for the open-ended classes. The total is $N = 121$ and cumulative counts are 15,35,71,111,121. The median lies in 100–150.For grouped data, use linear interpolation within the median class. Let M be the median, $L = 100$ its class's lower endpoint, $h = 50$ its width, $N = 121$ the total frequency, $C = 35$ the cumulative frequency before that class, and $f = 36$ its frequency. The median position is $N/2 = 60.5$; it is position 25.5 within the class. The equal interpolation step per observation is $h/f = 1.388889$.

$$M = L + h\frac{N/2 - C}{f} = 100 + 50\frac{60.5 - 35}{36} = 135.41667.$$

This estimates the median by assuming the class's observations are evenly spread; the exact ungrouped median cannot be recovered from grouped counts alone.

Q3(a). Standard deviation of marks

Question. Marks classes 0–10, ..., 90–100 have frequencies 1, 3, 6, 10, 12, 11, 6, 3, 2, 1. Find standard deviation.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	1	5	25
15	3	45	675
25	6	150	3750
35	10	350	12250
45	12	540	24300
55	11	605	33275
65	6	390	25350
75	3	225	16875
85	2	170	14450
95	1	95	9025

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 55$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{2575}{55} = 46.8182.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{139975}{55} - 46.8182^2 = 353.0579.$$

$$\sigma = \sqrt{\sigma^2} = 18.7898.$$

Q3(b). Pearson skewness

Question. Classes 0–10, ..., 70–80 have frequencies 5, 6, 11, 21, 35, 30, 22, 11. Calculate Karl Pearson skewness.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	5	25	125
15	6	90	1350
25	11	275	6875
35	21	735	25725
45	35	1575	70875
55	30	1650	90750
65	22	1430	92950
75	11	825	61875

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 141$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{6605}{141} = 46.8440.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{350525}{141} - 46.8440^2 = 291.6352.$$

$$\sigma = \sqrt{\sigma^2} = 17.0773.$$

The modal class has lower boundary $L = 40$ and width $h = 10$. Its frequency is $f_m = 35$; the preceding and following frequencies are $f_- = 21$ and $f_+ = 30$. The interpolated mode M_o is

$$M_o = L + h\frac{f_m - f_-}{2f_m - f_- - f_+} = 40 + 10\frac{14}{19} = 47.3684.$$

Karl Pearson's coefficient S_k measures asymmetry in standard-deviation units:

$$S_k = \frac{\bar{x} - M_o}{\sigma} = \frac{46.8440 - 47.3684}{17.0773} = -0.0307.$$

The distribution has negative (left) skewness. A coefficient near zero indicates little asymmetry; its sign describes the longer tail.

Q4(a). Normal income and percentiles

Question. For 3000 teachers, income is normal with mean 600 and SD 100. Find (i) percentage above 800; (ii) number below 400; (iii) lowest income in highest-paid 100; (iv) highest income among lowest-paid teachers; (v) three-standard-deviation range.

Worked solution. Let X denote the measured quantity, with population mean $\mu = 600$ and standard deviation $\sigma = 100$. Standardize using

$$Z = \frac{X - \mu}{\sigma}.$$

Here Z has a standard normal distribution, and $\Phi(z) = P(Z \le z)$ is its cumulative probability. Convert each endpoint separately, then subtract the cumulative probabilities.

$$P(X > 800) = 1 - \Phi(2.0000) = 0.022750.$$

$$P(X < 400) = \Phi(-2.0000) = 0.022750.$$

Thus (i) is 2.2750%; (ii) the expected number is $3000(0.0227501) = 68.25$, or about 68 teachers.

For (iii), let c be the threshold for the top 100 of 3,000. The cumulative fraction below it is $1 - 100/3000 = 0.966667$. With $z_r = \Phi^{-1}(r)$ denoting the standard-normal percentile at fraction r,

$$c = 600 + 100 z_{0.966667} = 783.3915.$$

For (iv), the paper omits the size or proportion of the lowest-paid group. If it contains m teachers, the cutoff is $600 + 100\Phi^{-1}(m/3000)$, where m must be specified. If the intended group was the bottom 100, it would be 416.6085; this is only a conditional illustration.

For (v), the interval within three standard deviations is $[\mu - 3\sigma, \mu + 3\sigma] = [300, 900]$, where $\mu = 600$ and $\sigma = 100$. Its width is 600 and it covers about 99.73% of the normal model.

Q4(b). Poisson job stream

Question. For mean two jobs per minute, find zero, exactly two, at most three, more than three, and the 90% upper count bound.

Worked solution. Let X be the event count in the stated interval, $\lambda = 2$ its mean count, k a nonnegative integer, and e the base of natural logarithms. With independent arrivals at a constant average rate,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-2}\frac{2^k}{k!}.$$

$$P(X = 0) = g(0) = 0.135335.$$

$$P(X = 2) = g(2) = 0.270671.$$

$$P(X \le 3) = \sum_{k=0}^{3} g(k) = 0.857123.$$

$$P(X > 3) = 1 - \sum_{k=0}^{3} g(k) = 0.142877.$$

For a 90% upper count bound, let m be the smallest integer with $P(X \le m) \ge 0.90$. Since $P(X \le 3) = 0.857123$ and $P(X \le 4) = 0.947347$, the bound is **4 jobs**. This is a 90% coverage bound, not an absolute maximum: a Poisson count has no finite maximum.

Q5(a). Conditional dice events

Question. A die is thrown twice. Find the probability of total 9 given points between 4 and 6, and the reverse conditional probability.

Worked solution. The phrase "the die has shown up points between 4 and 6" does not identify which throw or whether both throws are meant. State the event explicitly before counting.

A natural interpretation is that *both* throws lie in $\{4, 5, 6\}$. Let S mean sum 9 and D mean both throws are in that set. Nine ordered pairs satisfy D; only $(4, 5), (5, 4)$ also satisfy S. Four ordered pairs satisfy S: $(3, 6), (4, 5), (5, 4), (6, 3)$.

$$P(S \mid D) = \frac{2}{9}, \qquad P(D \mid S) = \frac{2}{4} = \frac{1}{2}.$$

If instead D refers only to the first throw, there are 18 outcomes in D and three in $D \cap S$; the answers are 1/6 and 3/4. If D means at least one throw lies in 4–6, its 27 outcomes include all four sum-9 outcomes, giving 4/27 and 1. The original wording does not select among these interpretations.

Q5(b). Economic growth given appreciation

Question. Growth levels high, moderate, low have probabilities 0.30, 0.50, 0.20. Rupee appreciation probabilities conditional on them are 0.70, 0.40, 0.20. Given appreciation, find the growth probabilities.

Worked solution. Let A denote appreciation, and H, M, L the three growth states. Their joint weights with appreciation are $0.30(0.70) = 0.21$, $0.50(0.40) = 0.20$, and $0.20(0.20) = 0.04$. Total probability gives $P(A) = 0.45$. Divide each joint weight by this total:

$$P(H \mid A) = \frac{0.21}{0.45} = 0.466667, \quad P(M \mid A) = \frac{0.20}{0.45} = 0.444444, \quad P(L \mid A) = \frac{0.04}{0.45} = 0.088889.$$

These posterior probabilities sum to one. High growth is the most probable state after observing appreciation, although it is not certain.

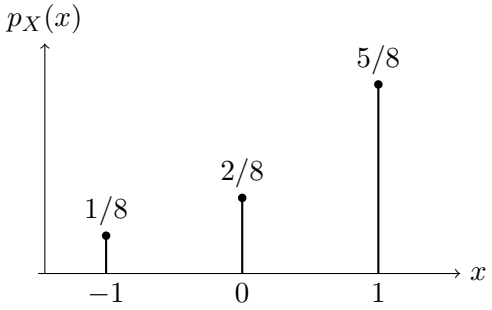
Q6(a). PMF and a squared variable

Question. X takes -1, 0, 1 with masses 1/8, 2/8, 5/8. Find E(X), plot its PMF, find the PMF and mean of Y=X squared, compute E(X squared) directly and Var(X).

Worked solution. Let $p_X(x)$ denote the probability mass at value x. The expectation $E(X)$ is the probability-weighted average and $E(X^2)$ its second raw moment:

$$E(X) = \sum_x x p_X(x) = (-1)(1/8) + 0 + 1(5/8) = 1/2.$$

Both $X = -1$ and $X = 1$ give $Y = X^2 = 1$. Therefore $P(Y = 0) = 2/8 = 1/4$ and $P(Y = 1) = 6/8 = 3/4$, with zero mass elsewhere. It follows that $E(Y) = 0(1/4) + 1(3/4) = 3/4$. Direct calculation confirms $E(X^2) = (-1)^2(1/8) + 0^2(2/8) + 1^2(5/8) = 3/4$. Thus $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 3/4 - 1/4 = 1/2$.



Q6(b). Independent events and complements

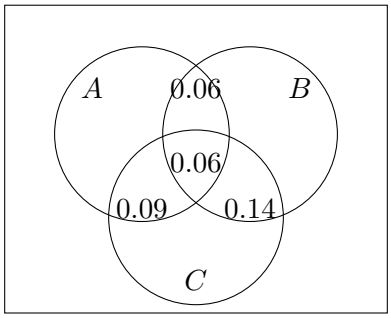
Question. Independent events have P(A)=0.3, P(B)=0.4, P(C)=0.5. Find $P(A \cap B \cap C^c)$, $P(A \cap B^c \cap C)$, and $P(A^c \cap B \cap C)$.

Worked solution. The superscript c denotes complement, so $P(A^c) = 0.7$, $P(B^c) = 0.6$ and $P(C^c) = 0.5$. Mutual independence also holds when events are replaced by their complements. Multiply the three appropriate factors:

$$P(A \cap B \cap C^c) = 0.3(0.4)(0.5) = 0.06,$$

$$P(A \cap B^c \cap C) = 0.3(0.6)(0.5) = 0.09,$$
$$P(A^c \cap B \cap C) = 0.7(0.4)(0.5) = 0.14.$$

These are the three pairwise-only regions of a Venn diagram, excluding its central triple intersection.



The centre is $P(A \cap B \cap C) = 0.3(0.4)(0.5) = 0.06$, different from the pairwise-only regions despite matching one numerical value.

Q7(a). Comparing colleges

Question. College A: course counts 200, 150, 100, with mean marks 70, 60, 80%. College B: counts 150, 100, 50 with means 80, 60, 80%. Which is better on average?

Worked solution. Use student counts as weights. Let n_i be students in course i, m_i its mean mark percentage, and $\bar{m} = \sum n_i m_i / \sum n_i$ the overall student-weighted mean.

$$\bar{m}_A = \frac{200(70) + 150(60) + 100(80)}{450} = 68.8889\%,$$
$$\bar{m}_B = \frac{150(80) + 100(60) + 50(80)}{300} = 73.3333\%.$$

College B has the higher overall average. A simple average of the three course averages would give each course equal weight regardless of enrolment and would answer a different question.

Q7(b). Testing errors

Question. Explain Type I and Type II errors.

Worked solution. The null hypothesis H_0 is the claim being tested; H_1 is the competing claim. A Type I error rejects H_0 when it is true. Its probability is α, the significance level. A Type II error fails to reject H_0 when a specified alternative is true. Its probability is β; the power $1 - \beta$ is the chance of detecting that alternative. For a bulb-lifespan test, Type I means declaring a change when the claimed mean is correct; Type II means missing a real change. Failure to reject is not proof of equality.

Decision	H_0 true	H_0 false
Reject H_0	Type I error	Correct detection
Do not reject H_0	Correct retention	Type II error

3 2023–24: Final / 2022–23 re-examination, 25 November 2023

Source P3. The cover identifies academic year 2023–24 and eligibility for the 2022–23 re-examination.

Q1(a). Classification

Question. Explain classification and list its types.

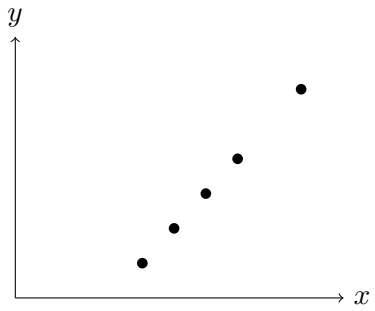
Worked solution. Classification groups observations according to a common characteristic so that the data become easier to compare. Categories should be clear, mutually exclusive, and collectively exhaustive.

- Qualitative classification uses attributes, such as employment sector or product type. Nominal categories have no natural order; ordinal categories, such as low, medium and high satisfaction, have an order.
- Quantitative classification uses numerical values. Discrete variables count items, such as defects; continuous variables measure quantities, such as waiting time. Numerical data may be grouped into class intervals.
- Chronological classification groups observations by time, for example annual sales.
- Geographical classification groups observations by location, for example district-wise rainfall.

Q1(b). Scatter plot

Question. Explain a scatter plot with a suitable example.

Worked solution. A scatter plot places one point for each observed pair (x_i, y_i), where x_i, y_i are the two measurements on item i. A rising pattern suggests positive association, a falling pattern negative association, and an unstructured cloud little linear association. A curved pattern may still indicate a strong non-linear relationship. The plot also reveals outliers. Correlation alone does not establish cause.



The illustrated pairs are the data in Q2(b). They lie on an upward straight line.

Q1(c). Vacation counts

Question. Among 20 independent workers, probability of two weeks vacation is 0.45; probability of three or more weeks is 0.20. Find (a) exactly 8 with two weeks, (b) at most 2 with three or more weeks.

Worked solution. Let X be the number of successes in $n = 20$ independent trials, each with success probability $p = 0.45$. The failure probability is $1 - p = 0.55$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 8) = b(8) = 0.162300.$$

For part (b), redefine success as three or more weeks, with count Y:Let X be the number of successes in $n = 20$ independent trials, each with success probability $p = 0.2$. The failure probability is $1 - p = 0.8$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(Y \leq 2) = \sum_{k=0}^{2} b(k) = 0.206085.$$

Q1(d). Expectation and variance transformations

Question. E(X)=10, Var(X)=3, E(Y)=20, Var(Y)=2. Find E(3Y-7), E(X-Y), Var(X-Y), Var(5X+9).

Worked solution. Let E denote expectation, Var variance and $\mathrm{Cov}(X, Y)$ covariance (the joint tendency to deviate from their means). For constants a, b, linearity gives $E(aY + b) = aE(Y) + b$, without any independence assumption. Variance obeys $\mathrm{Var}(aX+b) = a^2 \mathrm{Var}(X)$; adding a constant does not change spread. Thus

$$E(3Y - 7) = 3(20) - 7 = 53, \qquad E(X - Y) = 10 - 20 = -10,$$

$$\mathrm{Var}(5X + 9) = 5^2(3) = 75,$$

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y) = 5 - 2\,\mathrm{Cov}(X, Y).$$

The question supplies no covariance or independence condition. Therefore its numerical variance of $X - Y$ is undetermined. If X, Y are independent (or merely uncorrelated), covariance is zero and the answer is 5. Do not assume independence solely because separate means and variances are given.

Q2(a). Teacher driving times

Question. Driving-time classes 0–10, 10–20, 20–30, 30–40, 40–50 minutes have frequencies 3, 10, 6, 4, 2. Find the mean.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	3	15	75
15	10	150	2250
25	6	150	3750
35	4	140	4900
45	2	90	4050

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 25$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{545}{25} = 21.8000.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{15025}{25} - 21.8000^2 = 125.7600.$$

$$\sigma = \sqrt{\sigma^2} = 11.2143.$$

The requested estimated mean is 21.8 minutes; the spread calculations above also show the full grouped-data method.

Q2(b). Pearson correlation

Question. Pairs are (12, 2), (15, 4), (18, 6), (21, 8), (27, 12). Calculate correlation and interpret.

Worked solution. Let r be Pearson's sample correlation, $n = 5$ the number of pairs, and x_i, y_i the paired values. The calculation totals are $\sum x_i = 93$, $\sum y_i = 32$, $\sum x_i^2 = 1863$, $\sum y_i^2 = 264$, and $\sum x_i y_i = 684$.

$$r = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{[n\sum x_i^2 - (\sum x_i)^2][n\sum y_i^2 - (\sum y_i)^2]}}$$

$$= \frac{5(684) - 93(32)}{\sqrt{[5(1863) - 93^2][5(264) - 32^2]}} = \frac{444}{444} = 1.$$

There is perfect positive linear association: every pair satisfies $y = (2/3)x - 6$.

Q3(a). Boat production variation

Question. Daily production: 17, 21, 18, 27, 17, 21, 20, 22, 18, 23 boats. SD above 3 is considered unacceptable. Should the manager worry?

Worked solution. Let $n = 10$ be the sample size, x_i each day's output, $\bar{x}$ the sample mean, and s the sample standard deviation. Then $\bar{x} = 20.4$ and $\sum(x_i - \bar{x})^2 = 88.4$.

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n - 1}} = \sqrt{88.4/9} = 3.13404.$$

As a sample estimate, the SD exceeds 3, so it warrants attention under the manager's rule. If describing only these ten days using divisor n, the descriptive SD is $\sqrt{88.4/10} = 2.97321$, just below 3. The convention affects this borderline conclusion. No significance level is provided, so this is a comparison of descriptive estimates, not a formal population-variance test.

Q3(b). Red balls without replacement

Question. A bag has 3 red and 7 black balls. Given that the second ball is red, find the chance the first is also red.

Worked solution. Let R_1, R_2 denote red on the first and second draw. Without replacement, $P(R_1 \cap R_2) = (3/10)(2/9) = 1/15$. By symmetry, $P(R_2) = 3/10$.

$$P(R_1 \mid R_2) = \frac{P(R_1 \cap R_2)}{P(R_2)} = \frac{1/15}{3/10} = \frac{2}{9}.$$

Equivalently, after fixing a red ball in the second position, two red balls remain among the nine possible first-position balls.

Q4(a). White ball from three urns

Question. Urns contain white/black counts (3, 2), (2, 3), (4, 1). An urn is selected uniformly and a white ball observed. Find the chance it is urn 3.

Worked solution. Let U_i denote urn i and W a white ball. Each prior is $P(U_i) = 1/3$. The white likelihoods are $3/5, 2/5, 4/5$.

$$P(U_3 \mid W) = \frac{(1/3)(4/5)}{(1/3)(3/5 + 2/5 + 4/5)} = \frac{4}{9} = 0.444444.$$

The evidence favours urn 3 because it contains the greatest white fraction.

Q4(b). Running shoe lifetime

Question. Shoe lifetime is exponential with mean 18 months. Find probability it exceeds 15 months.

Worked solution. Let X be the waiting time, $\lambda = 1/18$ the rate per unit of time, and x a possible nonnegative duration. The exponential model has

$$f_X(x) = \lambda e^{-\lambda x}, \quad F_X(x) = 1 - e^{-\lambda x} \quad (x \geq 0).$$

Both functions are zero below zero. Here f_X is the density, F_X the CDF, and e the natural exponential base. Its mean and standard deviation are both $1/\lambda = 18$.

$$P(X > 15) = e^{-15/18} = 0.434598.$$

Q4(c). Uniform winter weight gain

Question. Weight gain is uniform from 0 to 30 pounds. Find probability between 10 and 15, expectation and SD.

Worked solution. Let X be weight gain, $a = 0, b = 30$ the endpoints, and $f_X(x) = 1/(b-a) = 1/30$ the density on this interval (zero outside). Probability is interval length divided by total length:

$$P(10 < X < 15) = \frac{15-10}{30-0} = \frac{1}{6}.$$

The expectation and standard deviation are

$$E(X) = \frac{a+b}{2} = 15, \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{30}{\sqrt{12}} = 8.66025 \text{ pounds}.$$

Here $E(X)$ is the population mean and σ its standard deviation.

Q5(a). Variance of a frequency table

Question. Values 10, 4, 6, 8 have frequencies 1, 3, 5, 1. Calculate variance.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
10	1	10	100
4	3	12	48
6	5	30	180
8	1	8	64

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 10$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{60}{10} = 6.0000.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{392}{10} - 6.0000^2 = 3.2000.$$

$$\sigma = \sqrt{\sigma^2} = 1.7889.$$

These are exact discrete values, so the descriptive variance is 3.2. The unbiased sample variance, if requested, is $3.2(10/9) = 3.55556$.

Q5(b). Fewer than two defectives

Question. 3% of units are defective. Among 200 independent units find probability of less than two defectives.

Worked solution. Let X be the number of successes in $n = 200$ independent trials, each with success probability $p = 0.03$. The failure probability is $1 - p = 0.97$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X < 2) = \sum_{k=0}^{1} b(k) = 0.016248.$$

A Poisson approximation uses mean $\lambda = np = 6$ and gives $P(X < 2) \approx e^{-6}(1 + 6) = 0.0173513$. Here λ is the mean defective count; the exact binomial result above is preferred unless approximation is requested.

Q6(a). Transforming a discrete variable

Question. X is uniform on $\{-1, 0, 1, 2\}$, with mass $1/4$ each. Find range and PMF of $Y = 3|X|$.

Worked solution. For a function $Y = g(X)$, apply g to every possible value of X; collect repeated outputs, then add the probabilities of all inputs producing each output:

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x).$$

Here x is an input, y an output, p_X, p_Y their PMFs, and $g(x) = 3|x|$. The inputs $-1, 0, 1, 2$ map to $3, 0, 3, 6$. Thus the range is $\{0, 3, 6\}$, with $P(Y = 0) = 1/4$, $P(Y = 3) = 1/2$, $P(Y = 6) = 1/4$. The output 3 has two preimages, so it must not receive probability $1/4$.

Q6(b). PMF and exponential PDF

Question. Explain a discrete PMF with an example and give the exponential PDF.

Worked solution. For a discrete random variable X, the probability mass function is $p_X(x) = P(X = x)$. Here x is a possible value; every mass is nonnegative and the masses sum to one. For a continuous X, the probability density function $f_X(x)$ is nonnegative and has total integral one. Probabilities are areas, not individual density heights:

$$P(a < X \le b) = \int_a^b f_X(u)\, du.$$

Here a, b are interval endpoints and u is the integration variable. A single point has probability zero in the continuous case. The cumulative distribution function is $F_X(x) = P(X \le x)$, so

$$F_X(x) = \sum_{u \le x} p_X(u) \quad \text{(discrete)}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(u)\, du \quad \text{(continuous)}.$$

Where the derivative exists, $f_X(x) = F_X'(x)$. For example, a fair die has mass $1/6$ at each integer from 1 to 6; a uniform measurement on $[0, 1]$ has density 1 inside that interval and CDF equal to 0 below 0, x between 0 and 1, and 1 above 1.Let X be the waiting time, $\lambda = 1/1$ the rate per unit of time, and x a possible nonnegative duration. The exponential model has

$$f_X(x) = \lambda e^{-\lambda x}, \quad F_X(x) = 1 - e^{-\lambda x} \quad (x \ge 0).$$

Both functions are zero below zero. Here f_X is the density, F_X the CDF, and e the natural exponential base. Its mean and standard deviation are both $1/\lambda = 1$.The last display uses unit mean as an example; for a general exponential distribution retain an arbitrary rate $\lambda > 0$.

Q7(a). Normally distributed burned area

Question. Annual burned area is normal with mean 4300 acres and SD 750 acres. Find probability of 2500–4200 acres.

Worked solution. Let X denote the measured quantity, with population mean $\mu = 4300$ and standard deviation $\sigma = 750$. Standardize using

$$Z = \frac{X - \mu}{\sigma}.$$

Here Z has a standard normal distribution, and $\Phi(z) = P(Z \leq z)$ is its cumulative probability. Convert each endpoint separately, then subtract the cumulative probabilities.

$$P(2500 < X < 4200) = \Phi(-0.1333) - \Phi(-2.4000) = 0.438767.$$

Q7(b). Two independent mean reductions

Question. High-tech firms: n=41, mean EPS reduction 13.8%, SD 18.9%. Consumer goods: n=35, mean 9.1%, SD 8.7%. At $\alpha = 0.10$, is the high-tech population mean reduction greater?

Worked solution. Let $n_1 = 41, n_2 = 35$ be sample sizes, $\bar{x}_1 = 13.8, \bar{x}_2 = 9.1$ the sample means, $s_1 = 18.9, s_2 = 8.7$ the sample standard deviations, and μ_1, μ_2 the population means. Assume independent random samples. Normal populations justify the t procedure for small samples; the large samples here also support the approximation for the sample means. Equal population variances are not stated, so use Welch's two-sample t-test.

Test $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 > 0$, at significance level $\alpha = 0.1$. The estimated standard error SE and t statistic are

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{18.9^2}{41} + \frac{8.7^2}{35}} = 3.297728,$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} = \frac{4.7}{3.297728} = 1.425224.$$

Let $v_1 = s_1^2/n_1 = 8.712439$ and $v_2 = s_2^2/n_2 = 2.162571$ denote the two estimated variance contributions. Welch's degrees of freedom ν are

$$\nu = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)} = 58.10975.$$

The upper-tail critical value is $t_{0.9,\nu} = 1.29629$, where $t_{0.9,\nu}$ is the 90th percentile of the t-distribution. Reject H_0 when t exceeds 1.29629. Since $1.425224 > 1.29629$, **reject** H_0. The data support a greater average EPS reduction for high-technology firms. If using a printed table, round degrees of freedom down to an available row and state the row used.

4 2023–24: Re-examination for 2022–23, 8 February 2024

Source P5. Academic year on the cover is 2023–24; the examination is labelled re-examination for 2022–23.

Q1(a). Classification and tabulation

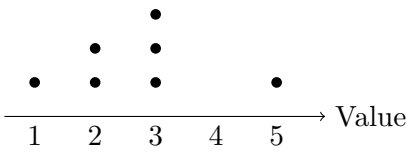
Question. Explain the difference between classification and tabulation.

Worked solution. Classification decides which groups the observations belong to; tabulation presents those groups and their counts or measurements in rows and columns. For example, separating workers into wage classes is classification; displaying the wage classes beside their frequencies is tabulation. Tabulation condenses a large dataset, enables comparison, exposes patterns and omissions, and supports calculation. A good table has a title, clear row and column headings, units, totals, and any necessary explanatory notes.

Q1(b). Dot plot

Question. Explain a dot plot with a suitable example.

Worked solution. A dot plot places one dot for each observation above its numerical value; repeated values stack vertically. It retains individual observations, making clusters, gaps and outliers easy to see. For the sample 1,2,2,3,3,3,5, the stack heights are 1,2,3,0,1 at values 1 through 5.



Q1(c). Busy telephones

Question. Each telephone is independently busy with probability 0.1. Of six selected numbers, find probability exactly four are busy.

Worked solution. Let X be the number of successes in $n = 6$ independent trials, each with success probability $p = 0.1$. The failure probability is $1 - p = 0.9$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 4) = b(4) = 0.001215.$$

Q1(d). Transformations of expectation and variance

Question. E(X)=15, Var(X)=4, E(Y)=30, Var(Y)=2. Find E(4Y-8), E(X-Y), Var(X-Y), Var(4X+8).

Worked solution. Let E denote expectation, Var variance and $\mathrm{Cov}(X, Y)$ covariance (the joint tendency to deviate from their means). For constants a, b, linearity gives $E(aY + b) = aE(Y) + b$, without any independence assumption. Variance obeys $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$; adding a constant does not change spread. Thus

$$E(4Y - 8) = 4(30) - 8 = 112, \qquad E(X - Y) = 15 - 30 = -15,$$

$$\mathrm{Var}(4X + 8) = 4^2(4) = 64,$$

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y) = 6 - 2\,\mathrm{Cov}(X, Y).$$

The question supplies no covariance or independence condition. Therefore its numerical variance of $X - Y$ is undetermined. If X, Y are independent (or merely uncorrelated), covariance is zero and the answer is 6. Do not assume independence solely because separate means and variances are given.

Q2(a). Mean by step-deviation

Question. Classes 20–29, ..., 120–129 have frequencies 6, 16, 21, 29, 25, 22, 11, 7, 4, 0, 2. Compute the sample mean by coding, assigning code 0 to 60–69. Do not make classes continuous.

Worked solution. Keep the class limits as printed and calculate each midpoint. Let $A = 64.5$ be the midpoint of 60–69, $h = 10$ the difference between consecutive midpoints, and $u_i = (m_i - A)/h$ the code for midpoint m_i. Let f_i be class frequency and $N = \sum f_i$ the total.

Class	Midpoint m_i	f_i	u_i	f_iu_i
20–29	24.5	6	-4	-24
30–39	34.5	16	-3	-48
40–49	44.5	21	-2	-42
50–59	54.5	29	-1	-29
60–69	64.5	25	0	0
70–79	74.5	22	1	22
80–89	84.5	11	2	22
90–99	94.5	7	3	21
100–109	104.5	4	4	16
110–119	114.5	0	5	0
120–129	124.5	2	6	12

The coding mean is

$$\bar{x} = A + h\frac{\sum f_iu_i}{N} = 64.5 + 10\frac{-50}{143} = 61.00350.$$

Here $\bar{x}$ is the sample's grouped mean. No continuity correction to class limits is needed for midpoint coding.

Q2(b). Spearman rank correlation

Question. X:10, 12, 8, 15, 20, 25, 40; Y:15, 10, 6, 25, 16, 12, 8. Find rank correlation and interpret.

Worked solution. Rank each variable in ascending order; there are no ties.

x_i	y_i	R_i	S_i	d_i	d_i^2
10	15	2	5	-3	9
12	10	3	3	0	0
8	6	1	1	0	0
15	25	4	7	-3	9
20	16	5	6	-1	1
25	12	6	4	2	4
40	8	7	2	5	25

Here R_i, S_i are ranks, $d_i = R_i - S_i$ their difference, and $n = 7$ is the number of pairs. Spearman's coefficient r_s is

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(48)}{7(49-1)} = 0.142857.$$

The sample shows very weak positive rank association; there is little monotonic relationship.

Q3(a). Bank workload variation

Question. Daily checks classes 0–199, 200–399, 400–599, 600–799, 800–999 have frequencies 10, 13, 17, 42, 18. Does SD exceed the operational threshold 200?

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
99.5	10	995	99002.5
299.5	13	3893.5	1.1661e+06
499.5	17	8491.5	4.2415e+06
699.5	42	29379	2.05506e+07
899.5	18	16191	1.45638e+07

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 100$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{58950}{100} = 589.5000.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{4.0621e+07}{100} - 589.5000^2 = 58700.0000.$$

$$\sigma = \sqrt{\sigma^2} = 242.2808.$$

The descriptive SD is about 251 checks, exceeding 200. Hank should therefore plan for staffing imbalance. Using the sample correction also gives a value above 200, so the operational conclusion does not depend on that convention.

Q3(b). A two-headed coin

Question. A box holds two fair coins and one two-headed coin. Pick one uniformly. (a) Find chance of heads. (b) Given heads, find chance the coin is two-headed.

Worked solution. Let D mean two-headed, F a fair coin and H heads. Then $P(D) = 1/3$, $P(F) = 2/3$, $P(H \mid D) = 1$ and $P(H \mid F) = 1/2$.

$$P(H) = P(D)P(H \mid D) + P(F)P(H \mid F) = 1/3 + (2/3)(1/2) = 2/3.$$

Bayes' rule gives $P(D \mid H) = [(1/3)(1)]/(2/3) = 1/2$. A head makes the two-headed coin more plausible, but does not identify it with certainty.

Q4(a). Selecting a box after seeing red

Question. Red/white counts in three boxes are (3, 2), (4, 5), (2, 4). Choose a box uniformly and observe red. Find probability of box 2.

Worked solution. Let B_i denote box i and R red. The red fractions are $3/5, 4/9, 1/3$ and the box priors are each $1/3$. Hence

$$P(B_2 \mid R) = \frac{(1/3)(4/9)}{(1/3)(3/5 + 4/9 + 1/3)} = \frac{20}{62} = \frac{10}{31} = 0.322581.$$

The common prior cancels; the likelihoods must still be normalized by their sum.

Q4(b). Uniform baseball duration

Question. Duration is uniform from 447 to 521 hours. Find mean and probability of 480–500 hours.

Worked solution. Let $a = 447, b = 521$ be the endpoints and X the duration. A uniform density is constant at $1/(b-a) = 1/74$ within the interval. Thus the population mean is $E(X) = (a+b)/2 = 484$ hours and

$$P(480 < X < 500) = \frac{500 - 480}{521 - 447} = \frac{10}{37} = 0.270270.$$

Q4(c). Waiting for a phone call

Question. Calls arrive at an average rate of 2 per hour. Find probability a call arrives within the next hour, expected waiting time and SD.

Worked solution. Assume a homogeneous Poisson arrival process; its interarrival waiting times are exponential. An average rate alone would not specify the waiting distribution.Let X be the waiting time, $\lambda = 1/0.5$ the rate per unit of time, and x a possible nonnegative duration. The exponential model has

$$f_X(x) = \lambda e^{-\lambda x}, \quad F_X(x) = 1 - e^{-\lambda x} \quad (x \geq 0).$$

Both functions are zero below zero. Here f_X is the density, F_X the CDF, and e the natural exponential base. Its mean and standard deviation are both $1/\lambda = 0.5$.Consequently $P(X \leq 1) = 1 - e^{-2} = 0.864665$. The expected wait and its SD are both 0.5 hour (30 minutes). If the question instead refers to the count N of calls in an hour, $E(N) = 2$ and its SD is $\sqrt{2}$; this is a different variable.

Q5(a). Grouped variance

Question. Classes 0–10, 10–20, 20–30, 30–40, 40–50 have frequencies 3, 6, 4, 2, 1. Find variance.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	3	15	75
15	6	90	1350
25	4	100	2500
35	2	70	2450
45	1	45	2025

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 16$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{320}{16} = 20.0000.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{8400}{16} - 20.0000^2 = 125.0000.$$

$$\sigma = \sqrt{\sigma^2} = 11.1803.$$

Q5(b). Funding elevator safety

Question. Accidents average five per week. Funds are allocated if probability of more than three accidents exceeds 70%. Will funds be allocated?

Worked solution. Let X be the event count in the stated interval, $\lambda = 5$ its mean count, k a nonnegative integer, and e the base of natural logarithms. With independent arrivals at a constant average rate,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-5}\frac{5^k}{k!}.$$

$$P(X > 3) = 1 - \sum_{k=0}^{3} g(k) = 0.734974.$$

Assuming a Poisson model, $P(X > 3) = 0.734974 > 0.70$, so the stated funding criterion is satisfied.

Q6(a). Squared transformation

Question. X has probability 1/4 at each of -1, 0, 1, 2. Find range and PMF of $Y = X^2$ and describe how it is obtained.

Worked solution. Map the four inputs to their squares: $-1 \mapsto 1$, $0 \mapsto 0$, $1 \mapsto 1$, $2 \mapsto 4$. The range is $\{0, 1, 4\}$. For any output y, add all input masses that map to it:

$$p_Y(y) = \sum_{x : x^2 = y} p_X(x).$$

Here p_X, p_Y are the input and output PMFs and the summation runs over all preimages x. Thus $P(Y = 0) = 1/4$, $P(Y = 1) = 1/2$, $P(Y = 4) = 1/4$, and all other values have zero probability.

Q6(b). Continuous PDF and CDF

Question. Describe a PDF and CDF for a continuous variable with an example.

Worked solution. For a discrete random variable X, the probability mass function is $p_X(x) = P(X = x)$. Here x is a possible value; every mass is nonnegative and the masses sum to one. For a continuous X, the probability density function $f_X(x)$ is nonnegative and has total integral one. Probabilities are areas, not individual density heights:

$$P(a < X \leq b) = \int_a^b f_X(u)\,du.$$

Here a, b are interval endpoints and u is the integration variable. A single point has probability zero in the continuous case. The cumulative distribution function is $F_X(x) = P(X \leq x)$, so

$$F_X(x) = \sum_{u \leq x} p_X(u) \quad \text{(discrete)}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(u)\,du \quad \text{(continuous)}.$$

Where the derivative exists, $f_X(x) = F_X'(x)$. For example, a fair die has mass $1/6$ at each integer from 1 to 6; a uniform measurement on $[0, 1]$ has density 1 inside that interval and CDF equal to 0 below 0, x between 0 and 1, and 1 above 1.

Q7(a). Normal pregnancy lengths

Question. Pregnancy length is normal, mean 266 days and SD 16 days. Find proportion lasting 240–270 days.

Worked solution. Let X denote the measured quantity, with population mean $\mu = 266$ and standard deviation $\sigma = 16$. Standardize using

$$Z = \frac{X - \mu}{\sigma}.$$

Here Z has a standard normal distribution, and $\Phi(z) = P(Z \leq z)$ is its cumulative probability. Convert each endpoint separately, then subtract the cumulative probabilities.

$$P(240 < X < 270) = \Phi(0.2500) - \Phi(-1.6250) = 0.546625.$$

Q7(b). Difference between two means

Question. Independent samples: n1=60, mean1=86, s1=6; n2=75, mean2=82, s2=9. Find estimated standard error and test equality of population means at $\alpha = 0.01$.

Worked solution. Let $n_1 = 60, n_2 = 75$ be sample sizes, $\bar{x}_1 = 86, \bar{x}_2 = 82$ the sample means, $s_1 = 6, s_2 = 9$ the sample standard deviations, and μ_1, μ_2 the population means. Assume independent random samples. Normal populations justify the t procedure for small samples; the large samples here also support the approximation for the sample means. Equal population variances are not stated, so use Welch's two-sample t-test.

Test $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$, at significance level $\alpha = 0.01$. The estimated standard error SE and t statistic are

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{6^2}{60} + \frac{9^2}{75}} = 1.296148,$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE} = \frac{4}{1.296148} = 3.086067.$$

Let $v_1 = s_1^2/n_1 = 0.600000$ and $v_2 = s_2^2/n_2 = 1.080000$ denote the two estimated variance contributions. Welch's degrees of freedom ν are

$$\nu = \frac{(v_1 + v_2)^2}{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)} = 129.08976.$$

The two-sided critical values are $\pm\, t_{0.995,\nu} = 2.61445$, where $t_{0.995,\nu}$ is the 99.5th percentile of the t-distribution. Reject H_0 when $|t|$ exceeds 2.61445. Since $3.086067 > 2.61445$, **reject** H_0. The data support different population means; the first sample has the higher observed mean. If using a printed table, round degrees of freedom down to an available row and state the row used.

5 2024–25: Final / earlier-batch re-examination, 11 December 2024

Source P6. Final for 2024–25; re-examination for 2022–23 and 2023–24.

Q1(a). Types of classification

Question. Explain the different types of classification.

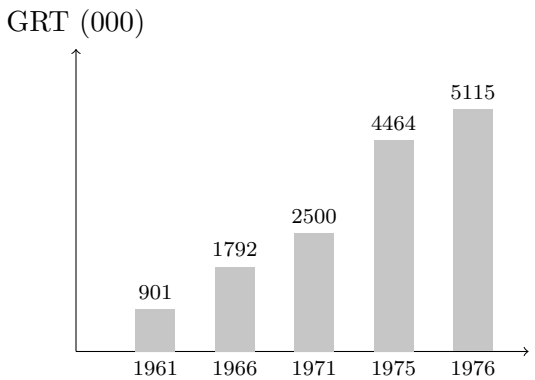
Worked solution. Classification groups observations according to a common characteristic so that the data become easier to compare. Categories should be clear, mutually exclusive, and collectively exhaustive.

- Qualitative classification uses attributes, such as employment sector or product type. Nominal categories have no natural order; ordinal categories, such as low, medium and high satisfaction, have an order.
- Quantitative classification uses numerical values. Discrete variables count items, such as defects; continuous variables measure quantities, such as waiting time. Numerical data may be grouped into class intervals.
- Chronological classification groups observations by time, for example annual sales.
- Geographical classification groups observations by location, for example district-wise rainfall.

Q1(b). Merchant fleet bar diagram

Question. Years 1961, 1966, 1971, 1975, 1976 have GRT (thousands) 901, 1792, 2500, 4464, 5115. Draw a suitable bar diagram.

Worked solution. Use separate equal-width bars with a common zero baseline. Place the years in order along the horizontal axis and GRT in thousands on the vertical axis. Bar heights, not bar widths, encode the measurements. The gaps emphasize that these are separate annual categories; spacing is categorical rather than proportional to elapsed time.



The plotted values show an increase across all the listed years; the largest tonnage is 5,115 thousand GRT in 1976.

Q1(c). Dice sums

Question. Two fair dice are thrown. Find probabilities of total five and total eight.

Worked solution. There are 36 equally likely ordered die pairs. Sum 5 occurs for $(1,4)$, $(2,3)$, $(3,2)$, $(4,1)$, so its probability is $4/36 = 1/9$. Sum 8 occurs for $(2,6)$, $(3,5)$, $(4,4)$, $(5,3)$, $(6,2)$, so its probability is $5/36$. Count ordered pairs because the two dice are distinguishable.

Q1(d). Probability functions

Question. Explain PMF, PDF, CDF and their relations.

Worked solution. For a discrete random variable X, the probability mass function is $p_X(x) = P(X = x)$. Here x is a possible value; every mass is nonnegative and the masses sum to one. For a continuous X, the probability density function $f_X(x)$ is nonnegative and has total integral one. Probabilities are areas, not individual density heights:

$$P(a < X \le b) = \int_a^b f_X(u)\,du.$$

Here a, b are interval endpoints and u is the integration variable. A single point has probability zero in the continuous case. The cumulative distribution function is $F_X(x) = P(X \le x)$, so

$$F_X(x) = \sum_{u \le x} p_X(u) \quad \text{(discrete)}, \qquad F_X(x) = \int_{-\infty}^{x} f_X(u)\,du \quad \text{(continuous)}.$$

Where the derivative exists, $f_X(x) = F_X'(x)$. For example, a fair die has mass $1/6$ at each integer from 1 to 6; a uniform measurement on $[0,1]$ has density 1 inside that interval and CDF equal to 0 below 0, x between 0 and 1, and 1 above 1.

Q2(a). Skewness in a bimodal distribution

Question. Marks classes 0–10, ..., 70–80 have frequencies 10, 40, 20, 0, 10, 40, 16, 14. Find Karl Pearson skewness and comment.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
5	10	50	250
15	40	600	9000
25	20	500	12500
35	0	0	0
45	10	450	20250
55	40	2200	121000
65	16	1040	67600
75	14	1050	78750

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 150$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{5890}{150} = 39.2667.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{309350}{150} - 39.2667^2 = 520.4622.$$
$$\sigma = \sqrt{\sigma^2} = 22.8136.$$

There are two equally high frequency peaks, at 10–20 and 50–60. A unique grouped mode cannot be chosen by the largest-frequency rule. State this issue, then use Pearson's second coefficient based on the median. Cumulative frequencies are 10,50,70,70,80,120,136,150. The median is in 40–50; its estimate is $M = 40 + 10(75 - 70)/10 = 45$.

Let $S_{k,2}$ denote Pearson's median-based coefficient, $\bar{x}$ the mean, M the median and σ the descriptive SD. Then

$$S_{k,2} = \frac{3(\bar{x} - M)}{\sigma} = \frac{3(39.266667 - 45)}{22.813641} = -0.753935.$$

The negative sign indicates left asymmetry by this measure. It does not remove the distribution's bimodality; a single skewness coefficient is not a full description of shape.

Q2(b). Median from class frequencies

Question. Classes 100–149.5, 150–199.5, 200–249.5, 250–299.5, 300–349.5, 350–399.5, 400–449.5, 450–499.5 have frequencies 12, 14, 27, 58, 72, 63, 36, 18. Find median class, position, interpolation step and median.

Worked solution. The total is 300 and cumulative counts are 12,26,53,111,183,246,282,300. Thus the 150th position is in 300–349.5. The printed class limits leave small gaps; use the intended consecutive 50-unit bands $[100, 150), [150, 200), \ldots$ for the main calculation.For grouped data, use linear interpolation within the median class. Let M be the median, $L = 300$ its class's lower endpoint, $h = 50$ its width, $N = 300$ the total frequency, $C = 111$ the cumulative frequency before that class, and $f = 72$ its frequency. The median position is $N/2 = 150$; it is position 39 within the class. The equal interpolation step per observation is $h/f = 0.694444$.

$$M = L + h\frac{N/2 - C}{f} = 300 + 50\frac{150 - 111}{72} = 327.08333.$$

This estimates the median by assuming the class's observations are evenly spread; the exact ungrouped median cannot be recovered from grouped counts alone.If measurements are rounded to the nearest 0.5 and continuous boundaries are reconstructed literally halfway between 299.5 and 300, use $L = 299.75$ and $h = 50$, obtaining 326.83333. State the boundary convention; the main result is 327.08333 under the usual 50-unit band interpretation.

Q3(a). Manufacturer and defect count

Question. Joint probabilities for manufacturer X at defects 0, 1, 2, 3 are 0.125, 0.0625, 0.1875, 0.125; for Y they are 0.0625, 0.0625, 0.125, 0.25. Find P(X given 0), P(0 given X), P(X given at least 2), P(Y given at least 1).

Worked solution. Let D be the defect count; X, Y denote manufacturer events. Row totals give $P(X) = P(Y) = 0.5$. Conditional probability divides the relevant joint probability by the probability of the condition.

$$P(X \mid D = 0) = \frac{0.125}{0.125 + 0.0625} = \frac{2}{3},$$
$$P(D = 0 \mid X) = \frac{0.125}{0.5} = \frac{1}{4},$$

$$P(X \mid D \geq 2) = \frac{0.1875 + 0.125}{0.1875 + 0.125 + 0.125 + 0.25} = \frac{5}{11},$$
$$P(Y \mid D \geq 1) = \frac{0.0625 + 0.125 + 0.25}{1 - (0.125 + 0.0625)} = \frac{7}{13}.$$

The condition determines the denominator; reversing it generally changes the answer.

Q3(b). Advertising source given purchase

Question. TV, radio and newspapers reach 30%, 50%, 20% respectively. Purchase probabilities given those sources are 0.6, 0.3, 0.1. A customer purchased. Find probability her source was radio.

Worked solution. Interpret the three media groups as mutually exclusive sources, as the paper intends. Let T, R, N denote the sources and B a purchase. The purchase probability is $P(B) = 0.3(0.6) + 0.5(0.3) + 0.2(0.1) = 0.35$. Therefore

$$P(R \mid B) = \frac{P(R)P(B \mid R)}{P(B)} = \frac{0.5(0.3)}{0.35} = \frac{3}{7} = 0.428571.$$

If source groups overlap in a real survey, these three percentages alone would not specify this partition model.

Q4(a). Binomial intervals

Question. $X \sim \text{Bin}(20, 0.4)$. Find P(X=6), P(X<9), P(X at least 12), and P(8 at most X at most 13).

Worked solution. Let X be the number of successes in $n = 20$ independent trials, each with success probability $p = 0.4$. The failure probability is $1 - p = 0.6$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 6) = b(6) = 0.124412.$$

$$P(X < 9) = \sum_{k=0}^{8} b(k) = 0.595599.$$

$$P(X \geq 12) = \sum_{k=12}^{20} b(k) = 0.056526.$$

$$P(8 \leq X \leq 13) = \sum_{k=8}^{13} b(k) = 0.577641.$$

The inequality in part (b) is marked unclearly in the scan. The result above uses the strict reading $X < 9$. If the intended symbol is $\leq$, include 9 as well: $P(X \leq 9) = 0.755337$.

Q4(b). Normalizing a discrete distribution

Question. Values -2, -1, 0, 1, 2, 3 have probabilities 0.2, k, 0.1, 2k, 0.1, 2k. Find k, mean and variance.

Worked solution. Let k be the unknown probability constant, X the variable and $p(x) = P(X = x)$ its mass function. Normalization gives $0.4 + 5k = 1$, so $k = 0.12$. Every mass is nonnegative. The expectation and second moment are

$$E(X) = \sum_x xp(x) = -2(0.2) - 0.12 + 0 + 0.24 + 2(0.1) + 3(0.24) = 0.64,$$

$$E(X^2) = \sum_x x^2 p(x) = 4(0.2) + 0.12 + 0 + 0.24 + 4(0.1) + 9(0.24) = 3.72.$$

The variance is $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 3.72 - 0.64^2 = 3.3104$. Here x runs over the six listed support values. The paper's term "density" should be read as PMF for this discrete variable.

Q5(a). Poisson approximation for screws

Question. 2% of screws are defective. Use Poisson to find probabilities among 100 screws of zero, one, at least two, and at most two defectives.

Worked solution. The approximation has mean $\lambda = np = 100(0.02) = 2$, where n is the screw count and p the defective fraction.Let X be the event count in the stated interval, $\lambda = 2$ its mean count, k a nonnegative integer, and e the base of natural logarithms. With independent arrivals at a constant average rate,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-2}\frac{2^k}{k!}.$$

$$P(X = 0) = g(0) = 0.135335.$$

$$P(X = 1) = g(1) = 0.270671.$$

$$P(X \geq 2) = 1 - \sum_{k=0}^{1} g(k) = 0.593994.$$

$$P(X \leq 2) = \sum_{k=0}^{2} g(k) = 0.676676.$$

Q5(b). First hit by a soldier

Question. Each independent shot hits with probability 0.7. Find first hit on shot 10, fewer than four shots, an even number of shots, and expected shots.

Worked solution. Let T be the shot number of the first hit, $p = 0.7$ the hit probability, $q = 1 - p = 0.3$ the miss probability, and k a positive integer. Independence gives the geometric law $P(T = k) = q^{k-1}p$: the first $k - 1$ shots must miss and shot k must hit.

$$P(T = 10) = 0.3^9(0.7) = 0.0000137781.$$

$$P(T < 4) = P(T \leq 3) = 1 - q^3 = 0.973.$$

For an even-numbered first hit, sum the geometric series:

$$P(T \text{ even}) = pq + pq^3 + pq^5 + \cdots = \frac{pq}{1 - q^2} = \frac{3}{13} = 0.230769.$$

The expected number of shots is $E(T) = 1/p = 10/7 = 1.428571$. If "hit on the 10th attempt" means simply that shot 10 hits, irrespective of earlier shots, that probability is 0.7; the sequence of subparts indicates that the intended interpretation is the first hit.

Q6(a). A two-variable discrete distribution

Question. $p(x, y) = k(2x+y)$ for x, y=0, 1, 2. Find k, marginals, independence, and conditional PMF of Y given X=x.

Worked solution. Although the paper calls this a density, the listed support is discrete, so it is a joint PMF. Let U, V denote the two variables (called X, Y or A, B in the question), and let $u, v \in \{0, 1, 2\}$. The constant k normalizes $p(u, v) = k(2u + v)$.

$$1 = k \sum_{u=0}^{2} \sum_{v=0}^{2} (2u + v) = 27k, \qquad k = 1/27.$$

The joint masses and row totals are:

$u\backslash v$	0	1	2	$p_U(u)$
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
$p_V(v)$	6/27	9/27	12/27	1

Marginalization means adding over the other variable:

$$p_U(u) = \sum_{v=0}^{2} p(u, v) = \frac{6u + 3}{27}, \qquad p_V(v) = \sum_{u=0}^{2} p(u, v) = \frac{6 + 3v}{27}.$$

Here p_U, p_V are the marginal PMFs. Since $p(0, 0) = 0$ but $p_U(0)p_V(0) = 2/81 > 0$, the variables are not independent. The conditional PMF of V given $U = u$ is

$$p_{V|U}(v \mid u) = \frac{p(u, v)}{p_U(u)} = \frac{2u + v}{6u + 3}, \qquad v = 0, 1, 2.$$

For every allowed u, these three probabilities sum to one.

Q6(b). Pooled confidence interval

Question. Independent normal samples with common variance have n1=4, mean1=12, SD1=4; n2=5, mean2=10, SD2=3. Find the 90% interval for the difference in population means.

Worked solution. Assume independent random samples from normal populations with equal variances. Let $n_1 = 4, n_2 = 5$ be the sample sizes, $\bar{x}_1 = 12, \bar{x}_2 = 10$ the sample means, and $s_1^2 = 16.00000, s_2^2 = 9.00000$ the unbiased sample variances. The pooled variance s_p^2, degrees of freedom ν, and standard error SE of the mean difference are

$$\nu = n_1 + n_2 - 2 = 7, \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{\nu} = 12.000000,$$

$$SE = \sqrt{s_p^2(1/n_1 + 1/n_2)} = 2.323790.$$

For a 90% confidence interval, the two-sided critical value is $t_{0.95,7} = 1.894579$. The population means are μ_1, μ_2, and their difference is estimated by $\bar{x}_1 - \bar{x}_2 = 2$:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.95,7} SE = 2 \pm 4.402603 = (-2.402603, 6.402603).$$

The stated confidence level refers to the long-run coverage of this interval procedure.The interval contains zero, so equality of population means remains compatible with these small samples at the corresponding two-sided 10% test level.

Q7(a). Green gram yield

Question. Expected yield is 12 quintals/hectare. Ten field yields are 14.3, 12.6, 13.7, 10.9, 13.7, 12.0, 11.4, 12.0, 12.6, 13.1. Do results conform?

Worked solution. The paper gives no significance level. Use and explicitly state the conventional two-sided 5% level.Use a one-sample t-test because the population standard deviation is unknown. Assume independent random observations from an approximately normal population. Let μ be the true mean, $\mu_0 = 12$ the claimed mean, $\bar{x}$ the sample mean, s the sample standard deviation, n the sample size, and $\nu = n - 1$ the degrees of freedom. Test $H_0 : \mu = 12$ against $H_1 : \mu \neq 12$.

$$n = 10, \qquad \bar{x} = 12.63000, \qquad \sum_{i=1}^{n}(x_i - \bar{x})^2 = 10.60100.$$

Here x_i is observation i. Therefore

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 1.08531, \qquad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = 1.83564.$$

At significance level $\alpha = 0.05$, $\nu = 9$ and the two-sided critical values are $\pm$ 2.26216. Do not reject H_0, since $|t|$ is smaller than the positive critical value. The observations are compatible with the claimed mean; this does not prove that it is exactly correct.At a different significance level the decision could change; state the level with the conclusion.

Q7(b). Bayes theorem

Question. State and prove Bayes theorem.

Worked solution. Let $B_1, \ldots, B_m$ be mutually exclusive and exhaustive possible sources, each with positive prior probability. Let E be observed evidence with $P(E) > 0$. Bayes' theorem states

$$P(B_j \mid E) = \frac{P(B_j)P(E \mid B_j)}{\sum_{i=1}^{m} P(B_i)P(E \mid B_i)}.$$

Here m is the number of sources, i indexes all sources, j identifies the source of interest, $P(B_j)$ is its prior probability, $P(E \mid B_j)$ is the likelihood of the evidence, and $P(B_j \mid E)$ is the revised or posterior probability. The denominator is $P(E)$.

For the proof, the definition of conditional probability gives $P(B_j \mid E) = P(B_j \cap E)/P(E)$. The multiplication rule gives $P(B_j \cap E) = P(B_j)P(E \mid B_j)$. Because the sources partition the sample space, the events $E \cap B_i$ are disjoint and their probabilities sum to $P(E)$. Substitution gives the displayed result. It is used to identify a likely source after observing an outcome, for example a defective product's factory or the box from which a coloured ball was drawn.

6 2024–25: Re-examination, 22 February 2025

Source P4. Re-examination for 2024–25 and the earlier 2022–23 and 2023–24 batches. The file contains six scanned pages despite the printed five-page count.

Q1(a). Tabulation

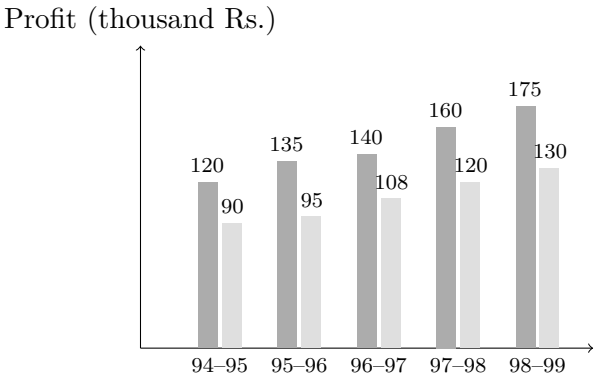
Question. Explain objectives of tabulation and distinguish it from classification.

Worked solution. Classification decides which groups the observations belong to; tabulation presents those groups and their counts or measurements in rows and columns. For example, separating workers into wage classes is classification; displaying the wage classes beside their frequencies is tabulation. Tabulation condenses a large dataset, enables comparison, exposes patterns and omissions, and supports calculation. A good table has a title, clear row and column headings, units, totals, and any necessary explanatory notes.

Q1(b). Comparative profit bars

Question. Years 1994–95, ..., 1998–99: company A profits 120, 135, 140, 160, 175; B profits 90, 95, 108, 120, 130 (thousands of rupees). Draw a suitable bar diagram.

Worked solution. Use a multiple bar diagram: two equal-width bars per year, a common zero baseline, and a legend distinguishing the companies.



Dark bars: Company A. Light bars: Company B.

Company A exceeds B in every listed year. Both profit sequences increase. The labels report the exact plotted heights.

Q1(c). Drawing coloured balls

Question. An urn has 8 white and 3 red balls. Draw two without replacement. Find probabilities both white, both red and one of each.

Worked solution. Let W, R denote the colours of each draw. There are $\binom{11}{2} = 55$ equally likely unordered pairs. With $\binom{n}{r}$ denoting the number of ways to choose r objects from n,

$$P(WW) = \frac{\binom{8}{2}}{55} = \frac{28}{55}, \qquad P(RR) = \frac{\binom{3}{2}}{55} = \frac{3}{55},$$

$$P(\text{one each}) = \frac{8(3)}{55} = \frac{24}{55}.$$

The three probabilities sum to one. Alternatively, for one of each add both draw orders: $(8/11)(3/10) + (3/11)(8/10)$.

Q1(d). Decision errors

Question. Explain Type I and Type II errors.

Worked solution. The null hypothesis H_0 is the claim being tested; H_1 is the competing claim. A Type I error rejects H_0 when it is true. Its probability is α, the significance level. A Type II error fails to reject H_0 when a specified alternative is true. Its probability is β; the power $1 - \beta$ is the chance of detecting that alternative. For a bulb-lifespan test, Type I means declaring a change when the claimed mean is correct; Type II means missing a real change. Failure to reject is not proof of equality.

Decision	H_0 true	H_0 false
Reject H_0	Type I error	Correct detection
Do not reject H_0	Correct retention	Type II error

Q2(a). Pearson skewness

Question. Classes 0–5, 5–10, ..., 35–40 have frequencies 2, 5, 7, 13, 21, 15, 8, 3. Find Karl Pearson skewness.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
2.5	2	5	12.5
7.5	5	37.5	281.25
12.5	7	87.5	1093.75
17.5	13	227.5	3981.25
22.5	21	472.5	10631.2
27.5	15	412.5	11343.8
32.5	8	260	8450
37.5	3	112.5	4218.75

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 74$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{1615}{74} = 21.8243.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{40012.5}{74} - 21.8243^2 = 64.4083.$$

$$\sigma = \sqrt{\sigma^2} = 8.0255.$$

The modal class has lower boundary $L = 20$ and width $h = 5$. Its frequency is $f_m = 21$; the preceding and following frequencies are $f_- = 13$ and $f_+ = 15$. The interpolated mode M_o is

$$M_o = L + h\frac{f_m - f_-}{2f_m - f_- - f_+} = 20 + 5\frac{8}{14} = 22.8571.$$

Karl Pearson's coefficient S_k measures asymmetry in standard-deviation units:

$$S_k = \frac{\bar{x} - M_o}{\sigma} = \frac{21.8243 - 22.8571}{8.0255} = -0.1287.$$

The distribution has negative (left) skewness. A coefficient near zero indicates little asymmetry; its sign describes the longer tail.

Q2(b). Median and interpolation

Question. Dollar classes 0–49.99, 50–99.99, ..., 450–499.99 have frequencies 78, 123, 187, 82, 51, 47, 13, 9, 6, 2 (total 600). Find median class, item number, interpolation step and median.

Worked solution. The cumulative counts begin 78,201,388, so the median class is 100–149.99. Use the intended 50-dollar bands.For grouped data, use linear interpolation within the median class. Let M be the median, $L = 100$ its class's lower endpoint, $h = 50$ its width, $N = 600$ the total frequency, $C = 201$ the cumulative frequency before that class, and $f = 187$ its frequency. The median position is $N/2 = 300$; it is position 99 within the class. The equal interpolation step per observation is $h/f = 0.267380$.

$$M = L + h\frac{N/2 - C}{f} = 100 + 50\frac{300 - 201}{187} = 126.47059.$$

This estimates the median by assuming the class's observations are evenly spread; the exact ungrouped median cannot be recovered from grouped counts alone.Using exact cent-based continuous boundaries would lower the result by 0.005 dollars, which does not affect the interpretation.

Q3(a). Completing an economist table

Question. 100 economists: academics stable 25 and contracting 20; private sector expanding 7, total 22; government stable 5, expanding 8, total 13. Stable total is 40. Complete the table and find P(A), P(E), P(A intersection E), P(G intersection C), P(E given G), P(G given E).

Worked solution. Let A, R, G denote academics, private sector and government; S, E, C denote stable, expanding and contracting. Government contracting is $13 - 5 - 8 = 0$. Private stable is $40 - 25 - 5 = 10$; private contracting is $22 - 10 - 7 = 5$. Academic total is $100 - 22 - 13 = 65$, leaving academic expanding $65 - 25 - 20 = 20$.

Sector	S	E	C	Total
A	25	20	20	65
R	10	7	5	22
G	5	8	0	13
Total	40	35	25	100

Divide each cell and margin by 100 to obtain the probability table. The requested probabilities are

$$P(A) = 0.65, \quad P(E) = 0.35, \quad P(A \cap E) = 0.20, \quad P(G \cap C) = 0,$$
$$P(E \mid G) = 8/13 = 0.615385, \qquad P(G \mid E) = 8/35 = 0.228571.$$

For the conditionals, divide by the row or column total of the event after the conditioning bar.

Q3(b). Cause of a fire

Question. Short-circuit and LPG causes have prior probabilities 0.8 and 0.2. Fire likelihoods are 0.30 and 0.95 respectively. Given fire, find the more probable cause.

Worked solution. Treat the two proposed causes as mutually exclusive and exhaustive, as intended by the supplied priors. Let S be short circuit, L LPG and F fire. Total fire probability is $P(F) = 0.8(0.30) + 0.2(0.95) = 0.43$. Therefore

$$P(S \mid F) = 0.24/0.43 = 24/43 = 0.558140, \qquad P(L \mid F) = 0.19/0.43 = 19/43 = 0.441860.$$

Short circuit is the more probable cause after observing the fire, despite LPG's higher conditional fire likelihood. This conclusion depends on the stated two-cause model.

Q4(a). Strict binomial inequalities

Question. $X \sim \text{Bin}(15, 0.35)$. Find P(X=8), P(X<7), P(X>9), P(5<X<10).

Worked solution. Let X be the number of successes in $n = 15$ independent trials, each with success probability $p = 0.35$. The failure probability is $1 - p = 0.65$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 8) = b(8) = 0.071037.$$

$$P(X < 7) = \sum_{k=0}^{6} b(k) = 0.754842.$$

$$P(X > 9) = \sum_{k=10}^{15} b(k) = 0.012443.$$

$$P(5 < X < 10) = \sum_{k=6}^{9} b(k) = 0.423275.$$

Q4(b). Normalizing a continuous density

Question. $f_X(x) = kx^2 e^{-x}$ for $x \geq 0$, zero otherwise. Find k, mean and variance.

Worked solution. Let k be the normalization constant and X the continuous variable. The integral identity $\int_0^\infty x^r e^{-x}\,dx = r!$ holds for a nonnegative integer r, by repeated integration by parts (boundary terms vanish). Normalization therefore gives

$$1 = k\int_0^\infty x^2 e^{-x}\,dx = 2k, \qquad k = 1/2.$$

The first and second raw moments are

$$E(X) = \int_0^\infty x f_X(x)\,dx = \tfrac{1}{2}(3!) = 3, \qquad E(X^2) = \int_0^\infty x^2 f_X(x)\,dx = \tfrac{1}{2}(4!) = 12.$$

Here E denotes expectation, x is the integration variable and $r!$ a factorial. The variance is $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 12 - 9 = 3$.

Q5(a). Poisson test-tube defects

Question. 2% of 100 test tubes are defective. Use Poisson to find zero, one, at least two and at most two defectives.

Worked solution. Set $\lambda = np = 100(0.02) = 2$, the expected defective count; n is the tube count and p the defect probability.Let X be the event count in the stated interval, $\lambda = 2$ its mean count, k a nonnegative integer, and e the base of natural logarithms. With independent arrivals at a constant average rate,

$$g(k) = P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} = e^{-2}\frac{2^k}{k!}.$$

$$P(X = 0) = g(0) = 0.135335.$$

$$P(X = 1) = g(1) = 0.270671.$$

$$P(X \geq 2) = 1 - \sum_{k=0}^{1} g(k) = 0.593994.$$

$$P(X \leq 2) = \sum_{k=0}^{2} g(k) = 0.676676.$$

Q5(b). Archer first-hit probabilities

Question. Hit probability per independent shot is 0.7. Find first hit on shot ten, fewer than four shots, even shot number, expected shots.

Worked solution. Let T be the shot number of the first hit, $p = 0.7$ the hit probability, $q = 1 - p = 0.3$ the miss probability, and k a positive integer. Independence gives the geometric law $P(T = k) = q^{k-1}p$: the first $k - 1$ shots must miss and shot k must hit.

$$P(T = 10) = 0.3^9(0.7) = 0.0000137781.$$

$$P(T < 4) = P(T \leq 3) = 1 - q^3 = 0.973.$$

For an even-numbered first hit, sum the geometric series:

$$P(T \text{ even}) = pq + pq^3 + pq^5 + \cdots = \frac{pq}{1 - q^2} = \frac{3}{13} = 0.230769.$$

The expected number of shots is $E(T) = 1/p = 10/7 = 1.428571$. If "hit on the 10th attempt" means simply that shot 10 hits, irrespective of earlier shots, that probability is 0.7; the sequence of subparts indicates that the intended interpretation is the first hit.

Q6(a). Joint distribution on nine points

Question. $p(a,b) = k(2a+b)$, a, b=0, 1, 2. Find k, marginals, independence and conditional law of B given A=a.

Worked solution. Although the paper calls this a density, the listed support is discrete, so it is a joint PMF. Let U, V denote the two variables (called X, Y or A, B in the question), and let $u, v \in \{0,1,2\}$. The constant k normalizes $p(u,v) = k(2u+v)$.

$$1 = k\sum_{u=0}^{2}\sum_{v=0}^{2}(2u+v) = 27k, \qquad k = 1/27.$$

The joint masses and row totals are:

$u\backslash v$	0	1	2	$p_U(u)$
0	0	1/27	2/27	3/27
1	2/27	3/27	4/27	9/27
2	4/27	5/27	6/27	15/27
$p_V(v)$	6/27	9/27	12/27	1

Marginalization means adding over the other variable:

$$p_U(u) = \sum_{v=0}^{2} p(u,v) = \frac{6u+3}{27}, \qquad p_V(v) = \sum_{u=0}^{2} p(u,v) = \frac{6+3v}{27}.$$

Here p_U, p_V are the marginal PMFs. Since $p(0,0) = 0$ but $p_U(0)p_V(0) = 2/81 > 0$, the variables are not independent. The conditional PMF of V given $U = u$ is

$$p_{V|U}(v \mid u) = \frac{p(u,v)}{p_U(u)} = \frac{2u+v}{6u+3}, \qquad v = 0,1,2.$$

For every allowed u, these three probabilities sum to one.

Q6(b). Medication and IQ

Question. Population benchmark IQ is 100. Thirty medication recipients have mean 140 and sample SD 20. Test any change at 5%.

Worked solution. Let μ be mean IQ under the medication, $\mu_0 = 100$ the benchmark, $\bar{x} = 140$ the sample mean, $s = 20$ sample SD, and $n = 30$ the sample size. Test $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$, assuming independent approximately normal measurements.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{40}{20/\sqrt{30}} = 10.95445.$$

The degrees of freedom are $\nu = 29$. At two-sided $\alpha = 0.05$, critical values are $\pm t_{0.975,29} = \pm 2.04523$. Since $|t| > 2.04523$, reject equality: the treated sample supports a different, higher mean than the benchmark. A comparison without a randomized control group does not by itself isolate a causal medication effect.

Q7(a). Bayes theorem and proof

Question. State Bayes theorem and explain it with proof.

Worked solution. Let $B_1, \ldots, B_m$ be mutually exclusive and exhaustive possible sources, each with positive prior probability. Let E be observed evidence with $P(E) > 0$. Bayes' theorem states

$$P(B_j \mid E) = \frac{P(B_j)P(E \mid B_j)}{\sum_{i=1}^{m} P(B_i)P(E \mid B_i)}.$$

Here m is the number of sources, i indexes all sources, j identifies the source of interest, $P(B_j)$ is its prior probability, $P(E \mid B_j)$ is the likelihood of the evidence, and $P(B_j \mid E)$ is the revised or posterior probability. The denominator is $P(E)$.

For the proof, the definition of conditional probability gives $P(B_j \mid E) = P(B_j \cap E)/P(E)$. The multiplication rule gives $P(B_j \cap E) = P(B_j)P(E \mid B_j)$. Because the sources partition the sample space, the events $E \cap B_i$ are disjoint and their probabilities sum to $P(E)$. Substitution gives the displayed result. It is used to identify a likely source after observing an outcome, for example a defective product's factory or the box from which a coloured ball was drawn.

Q7(b). Pooled variance, interval and test

Question. Samples from two independent normal populations: first 12, 3, 8, 5; second 14, 7, 7, 9, 6. Find pooled variance, 95% interval for mean difference and test equality at 5%.

Worked solution. Calculate each sample mean and unbiased sample variance using $s_j^2 = \sum_i (x_{ji} - \bar{x}_j)^2/(n_j - 1)$, where j labels the sample, i its observations, n_j its size, and $\bar{x}_j$ its mean. The two sums of squared deviations are 46 and 41.2, respectively. As a pooled analysis, assume equal population variances.Assume independent random samples from normal populations with equal variances. Let $n_1 = 4, n_2 = 5$ be the sample sizes, $\bar{x}_1 = 7, \bar{x}_2 = 8.6$ the sample means, and $s_1^2 = 15.33333, s_2^2 = 10.30000$ the unbiased sample variances. The pooled variance s_p^2, degrees of freedom ν, and standard error SE of the mean difference are

$$\nu = n_1 + n_2 - 2 = 7, \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{\nu} = 12.457143,$$

$$SE = \sqrt{s_p^2(1/n_1 + 1/n_2)} = 2.367639.$$

For a 95% confidence interval, the two-sided critical value is $t_{0.975,7} = 2.364624$. The population means are μ_1, μ_2, and their difference is estimated by $\bar{x}_1 - \bar{x}_2 = -1.6$:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{0.975,7} SE = -1.6 \pm 5.598577 = (-7.198577, 3.998577).$$

The stated confidence level refers to the long-run coverage of this interval procedure.For $H_0 : \mu_1 - \mu_2 = 0$ against $H_1 : \mu_1 - \mu_2 \neq 0$, the statistic is $t = (7 - 8.6)/\sqrt{(87.2/7)(1/4 + 1/5)} = -0.67578$ approximately. Since $|t| < 2.36462$ with 7 degrees of freedom, do not reject H_0. The samples do not provide sufficient evidence of different means at 5%; they do not establish equality.

7 2025–26: Final examination, 29 November 2025

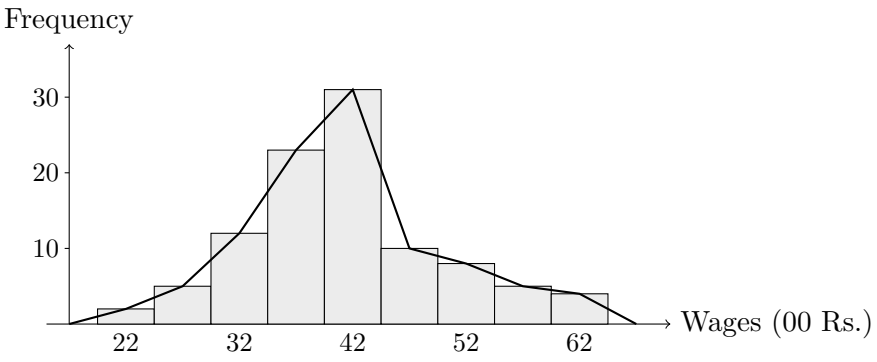
Sources P2 and P8 are byte-for-byte identical. This chapter solves their common paper once in full.

Q1(a). Histogram and frequency polygon

Question. Weekly wages in hundreds of rupees: 20–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50–54, 55–59, 60–64 have frequencies 2, 5, 12, 23, 31, 10, 8, 5, 4. Draw histogram and frequency polygon.

Worked solution. The frequencies sum to 100 workers. The integer wage-class labels become continuous boundaries 19.5–24.5,24.5–29.5,...,59.5–64.5. Each class width is 5, so frequency heights can be used directly (a frequency-density histogram would divide every height by 5). Adjacent rectangles touch.

The midpoint coordinates for the polygon are $(22, 2)$, $(27, 5)$, $(32, 12)$, $(37, 23)$, $(42, 31)$, $(47, 10)$, $(52, 8)$, $(57, 5)$, $(62, 4)$. Add $(17, 0), (67, 0)$ to close the polygon.



The diagram overlays the polygon on the histogram for comparison. The modal wage class is 40–44.

Q1(b). Binomial versus Poisson

Question. Compare binomial and Poisson distributions.

Worked solution. The binomial counts successes in a fixed number n of independent trials, each with success probability p. The Poisson counts occurrences in a fixed exposure interval, with mean count λ, typically assuming independent events and a constant rate.

Property	Binomial	Poisson
Support	$0, 1, \ldots, n$	$0, 1, 2, \ldots$
Parameters	n, p	$\lambda > 0$
PMF	$\binom{n}{k} p^k (1-p)^{n-k}$	$e^{-\lambda}\lambda^k / k!$
Mean	np	λ
Variance	$np(1-p)$	λ

Here k is the event count, e is the exponential base, and the PMF is the probability of that count. A Poisson approximation to binomial uses $\lambda = np$ when n is large and p small. For example, the

count of defects in 100 items with defect probability 0.02 may be approximated by Poisson mean 2. The two models are not interchangeable merely because both count events.

Q1(c). One white and one black

Question. A bag holds 7 white and 9 black balls. Two are drawn successively without replacement. Find probability of one of each colour.

Worked solution. Let W, B denote white and black draw outcomes. The desired event is the disjoint union of orders WB and BW. Thus

$$P(\text{one each}) = \frac{7}{16}\frac{9}{15} + \frac{9}{16}\frac{7}{15} = \frac{126}{240} = \frac{21}{40} = 0.525.$$

The second denominator is 15 because the first ball is not replaced. Counting only the first order would miss half of the favourable cases.

Q1(d). Qualitative and quantitative classification

Question. Explain qualitative and quantitative classification in detail.

Worked solution. Classification groups observations according to a common characteristic so that the data become easier to compare. Categories should be clear, mutually exclusive, and collectively exhaustive.

- Qualitative classification uses attributes, such as employment sector or product type. Nominal categories have no natural order; ordinal categories, such as low, medium and high satisfaction, have an order.
- Quantitative classification uses numerical values. Discrete variables count items, such as defects; continuous variables measure quantities, such as waiting time. Numerical data may be grouped into class intervals.
- Chronological classification groups observations by time, for example annual sales.
- Geographical classification groups observations by location, for example district-wise rainfall.

For quantitative grouped classes, use a consistent endpoint rule, such as $[20, 25)$ meaning 20 included and 25 excluded, to prevent double counting. For qualitative data, frequencies can summarize categories even when averages of category labels have no meaning.

Q2(a). Skewness of wages

Question. Wage classes 70–80, ..., 140–150 have frequencies 12, 18, 35, 42, 50, 45, 20, 8. Calculate Karl Pearson skewness and explain it.

Worked solution. Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
75	12	900	67500
85	18	1530	130050
95	35	3325	315875
105	42	4410	463050
115	50	5750	661250
125	45	5625	703125
135	20	2700	364500
145	8	1160	168200

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 230$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{25400}{230} = 110.4348.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{2.87355e+06}{230} - 110.4348^2 = 297.8544.$$

$$\sigma = \sqrt{\sigma^2} = 17.2585.$$

The modal class has lower boundary $L = 110$ and width $h = 10$. Its frequency is $f_m = 50$; the preceding and following frequencies are $f_- = 42$ and $f_+ = 45$. The interpolated mode M_o is

$$M_o = L + h\frac{f_m - f_-}{2f_m - f_- - f_+} = 110 + 10\frac{8}{13} = 116.1538.$$

Karl Pearson's coefficient S_k measures asymmetry in standard-deviation units:

$$S_k = \frac{\bar{x} - M_o}{\sigma} = \frac{110.4348 - 116.1538}{17.2585} = -0.3314.$$

The distribution has negative (left) skewness. A coefficient near zero indicates little asymmetry; its sign describes the longer tail.

Q2(b). Bayes theorem: statement and proof

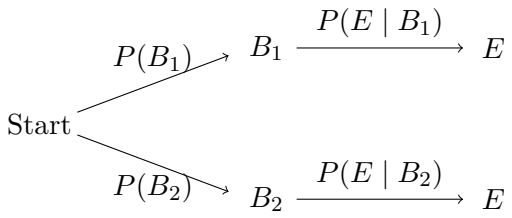
Question. State Bayes theorem; explain its components, proof and applications.

Worked solution. Let $B_1, \ldots, B_m$ be mutually exclusive and exhaustive possible sources, each with positive prior probability. Let E be observed evidence with $P(E) > 0$. Bayes' theorem states

$$P(B_j \mid E) = \frac{P(B_j)P(E \mid B_j)}{\sum_{i=1}^{m} P(B_i)P(E \mid B_i)}.$$

Here m is the number of sources, i indexes all sources, j identifies the source of interest, $P(B_j)$ is its prior probability, $P(E \mid B_j)$ is the likelihood of the evidence, and $P(B_j \mid E)$ is the revised or posterior probability. The denominator is $P(E)$.

For the proof, the definition of conditional probability gives $P(B_j \mid E) = P(B_j \cap E)/P(E)$. The multiplication rule gives $P(B_j \cap E) = P(B_j)P(E \mid B_j)$. Because the sources partition the sample space, the events $E \cap B_i$ are disjoint and their probabilities sum to $P(E)$. Substitution gives the displayed result. It is used to identify a likely source after observing an outcome, for example a defective product's factory or the box from which a coloured ball was drawn.A probability tree organizes the calculation: multiply along a branch to get a joint probability, add evidence branches to obtain the denominator, and divide the selected branch by that denominator.



Only the evidence branches are shown. Each source also has a complementary branch for E^c, the absence of the evidence.

Q3. Missing frequencies and descriptive measures

Question. Classes 100–110, ..., 190–200 have frequencies 4, 7, 15, f1, 40, f2, 16, 10, 6, 3. Total 150, median 146.25. Find f1, f2, mean, SD and mode.

Worked solution. Let f_1, f_2 denote the unknown class counts. The known counts sum to 101, so $f_1 + f_2 = 49$. The median class is 140–150, with lower boundary $L = 140$, width $h = 10$, frequency 40 and preceding cumulative frequency $26 + f_1$. Median interpolation gives

$$146.25 = 140 + 10\frac{75 - (26 + f_1)}{40}, \quad 49 - f_1 = 25.$$

Hence $f_1 = 24$, $f_2 = 25$. These frequencies sum correctly to 150. Now use the completed table for the mean, SD and mode.Use the class midpoints as representative values. This gives an estimate because the individual observations inside each class are unavailable.

Midpoint m_i	Frequency f_i	$f_i m_i$	$f_i m_i^2$
105	4	420	44100
115	7	805	92575
125	15	1875	234375
135	24	3240	437400
145	40	5800	841000
155	25	3875	600625
165	16	2640	435600
175	10	1750	306250
185	6	1110	205350
195	3	585	114075

Here m_i is midpoint of class i, f_i is its frequency, $N = \sum f_i = 150$ is the total count, $\bar{x}$ is the estimated mean, σ^2 is the descriptive variance, and σ is the standard deviation.

$$\bar{x} = \frac{\sum f_i m_i}{N} = \frac{22100}{150} = 147.3333.$$

$$\sigma^2 = \frac{\sum f_i m_i^2}{N} - \bar{x}^2 = \frac{3.31135e+06}{150} - 147.3333^2 = 368.5556.$$

$$\sigma = \sqrt{\sigma^2} = 19.1978.$$

The modal class has lower boundary $L = 140$ and width $h = 10$. Its frequency is $f_m = 40$; the preceding and following frequencies are $f_- = 24$ and $f_+ = 25$. The interpolated mode M_o is

$$M_o = L + h\frac{f_m - f_-}{2f_m - f_- - f_+} = 140 + 10\frac{16}{31} = 145.1613.$$

Karl Pearson's coefficient S_k measures asymmetry in standard-deviation units:

$$S_k = \frac{\bar{x} - M_o}{\sigma} = \frac{147.3333 - 145.1613}{19.1978} = 0.1131.$$

The distribution has positive (right) skewness. A coefficient near zero indicates little asymmetry; its sign describes the longer tail.The question asks for the mode but not skewness; the final coefficient above additionally shows how this question connects to Q2(a).

Q4(a). Joint PMF with squared term

Question. $p(x, y) = k(x^2 + y)$ for x=1, 2 and y=1, 2, 3. Find k, marginals, independence and conditional law of X given Y=2.

Worked solution. Let $p(x, y)$ be the joint probability mass and k its normalizing constant. Summing the six weights gives $2 + 3 + 4 + 5 + 6 + 7 = 27$, hence $k = 1/27$.

$x\backslash y$	1	2	3	$p_X(x)$
1	2/27	3/27	4/27	9/27
2	5/27	6/27	7/27	18/27
$p_Y(y)$	7/27	9/27	11/27	1

The marginals p_X, p_Y are row and column sums:

$$p_X(x) = \frac{3x^2 + 6}{27} \ (x = 1, 2), \qquad p_Y(y) = \frac{5 + 2y}{27} \ (y = 1, 2, 3).$$

Independence fails because $p(1, 1) = 2/27 = 6/81$ while $p_X(1)p_Y(1) = (1/3)(7/27) = 7/81$. For the conditional distribution,

$$p_{X|Y}(x \mid 2) = \frac{p(x, 2)}{p_Y(2)} = \frac{x^2 + 2}{9}.$$

Therefore the probabilities at $X = 1, 2$ given $Y = 2$ are $1/3, 2/3$ respectively; they sum to one.

Q4(b). Ten fair coins

Question. Ten unbiased coins are tossed independently. Find probabilities of exactly 6 heads, at least 8, no heads, at least one, at most 3, and at least 4.

Worked solution. Let X be the number of successes in $n = 10$ independent trials, each with success probability $p = 0.5$. The failure probability is $1 - p = 0.5$. For an integer count k,

$$b(k) = P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad \binom{n}{k} = \frac{n!}{k!(n - k)!}.$$

Here $b(k)$ is the probability of exactly k successes and the factorial counts arrangements. For an interval, add $b(k)$ over the permitted integers; pay particular attention to strict inequalities.

$$P(X = 6) = b(6) = 0.205078.$$

$$P(X \geq 8) = \sum_{k=8}^{10} b(k) = 0.054688.$$

$$P(X = 0) = b(0) = 0.000977.$$

$$P(X \geq 1) = \sum_{k=1}^{10} b(k) = 0.999023.$$

$$P(X \leq 3) = \sum_{k=0}^{3} b(k) = 0.171875.$$

$$P(X \geq 4) = \sum_{k=4}^{10} b(k) = 0.828125.$$

Because all $2^{10} = 1024$ head-tail sequences are equally likely, these answers can also be written as 210/1024, 56/1024, 1/1024, 1023/1024, 176/1024, 848/1024.

Q5(a). Exponential call-centre wait

Question. Waiting time has exponential mean 5 minutes. Find PDF, CDF, probability above 8 minutes and probability between 3 and 7 minutes.

Worked solution. Let X be the waiting time, $\lambda = 1/5$ the rate per unit of time, and x a possible nonnegative duration. The exponential model has

$$f_X(x) = \lambda e^{-\lambda x}, \quad F_X(x) = 1 - e^{-\lambda x} \quad (x \geq 0).$$

Both functions are zero below zero. Here f_X is the density, F_X the CDF, and e the natural exponential base. Its mean and standard deviation are both $1/\lambda = 5$.

$$P(X > 8) = e^{-8/5} = 0.201897.$$

For the interval, subtract the two cumulative probabilities: $P(3 < X < 7) = e^{-3/5} - e^{-7/5} = 0.302215$. The density is zero for negative waits and integrates to one over all nonnegative waits.

Q5(b). Red ball from box 3

Question. Boxes 1, 2, 3 have red/green counts (5, 5), (2, 8), (7, 3). Choose a box uniformly and observe red. Find probability of box 3.

Worked solution. Let B_i mean box i and R red. All priors equal 1/3, while the red likelihoods are $0.5, 0.2, 0.7$.

$$P(B_3 \mid R) = \frac{(1/3)(0.7)}{(1/3)(0.5 + 0.2 + 0.7)} = \frac{0.7}{1.4} = \frac{1}{2}.$$

The denominator $P(R) = 1.4/3$ includes red outcomes from every box; omitting other sources would overstate certainty.

Q6(a). Testing errors and distribution shape

Question. Explain Type I and II errors, skewness and kurtosis in detail.

Worked solution. The null hypothesis H_0 is the claim being tested; H_1 is the competing claim. A Type I error rejects H_0 when it is true. Its probability is α, the significance level. A Type II error fails to reject H_0 when a specified alternative is true. Its probability is β; the power $1-\beta$ is the chance of detecting that alternative. For a bulb-lifespan test, Type I means declaring a change when the claimed mean is correct; Type II means missing a real change. Failure to reject is not proof of equality.

Decision	H_0 true	H_0 false
Reject H_0	Type I error	Correct detection
Do not reject H_0	Correct retention	Type II error

Skewness describes asymmetry. A positive value usually indicates a longer right tail; a negative value a longer left tail. A moment coefficient is $\gamma_1 = E[(X-\mu)^3]/\sigma^3$, where X is the variable, $\mu = E(X)$ its mean, σ its positive standard deviation and E denotes expectation. A symmetric distribution has zero third central moment when it exists; zero skewness need not imply symmetry.

Kurtosis describes standardized fourth-moment behaviour, particularly sensitivity to tails and extreme observations. Define $\beta_2 = E[(X-\mu)^4]/\sigma^4$, with excess kurtosis $\gamma_2 = \beta_2 - 3$. The normal distribution has $\beta_2 = 3$ and $\gamma_2 = 0$ (mesokurtic). Values above 3 are termed leptokurtic and values below 3 platykurtic. Kurtosis is not solely a measure of the visible height of a peak, and equal kurtosis need not mean equal shape.

Q6(b). Fitting a Poisson model

Question. Error counts per hour 0, 1, 2, 3, 4 have observed frequencies 120, 55, 18, 5, 2 over 200 hours. Find theoretical expected frequencies, mean and variance.

Worked solution. Let x be the hourly error count, f_x its observed frequency, and $N = 200$ the number of testing hours. Fit the Poisson mean by the observed average:

$$\widehat{\lambda} = \frac{\sum_x x f_x}{N} = \frac{55 + 2(18) + 3(5) + 4(2)}{200} = \frac{114}{200} = 0.57.$$

For count k, the expected frequency E_k is N times its fitted probability:

$$E_k = N e^{-\widehat{\lambda}} \frac{\widehat{\lambda}^k}{k!}.$$

Here $k!$ is the factorial; expected counts need not be integers.

Count k	Observed	Expected E_k
0	120	113.105088
1	55	64.469900
2	18	18.373922
3	5	3.491045
4	2	0.497474
≥ 5	0	0.062572

Retain the ≥ 5 tail: truncating at 4 would lose probability mass and make the expected counts sum to less than 200. The fitted distribution's theoretical mean and variance are both $\widehat{\lambda} = 0.57$; this does not require the empirical variance to be exactly 0.57. A goodness-of-fit test is not requested.

Q7(a). A bounded power density

Question. $f_X(x) = a/x^5$ for $2 \le x \le 10$, zero otherwise. Find a, the CDF, and $P(4 \le X \le 7)$.

Worked solution. Let a be the normalization constant, x a possible value, and u an integration variable. Since total probability is one,

$$1 = a\int_2^{10} x^{-5}\,dx = \frac{a}{4}(2^{-4} - 10^{-4}), \qquad a = \frac{4}{1/16 - 1/10000} = \frac{2500}{39} = 64.102564.$$

Let $F_X(x) = P(X \le x)$ be the cumulative distribution function. Integrating from the lower support limit gives

$$F_X(x) = \begin{cases} 0, & x < 2, \\ \dfrac{a}{4}(2^{-4} - x^{-4}), & 2 \le x \le 10, \\ 1, & x > 10. \end{cases}$$

The interval probability is

$$P(4 \le X \le 7) = F_X(7) - F_X(4) = \frac{a}{4}(4^{-4} - 7^{-4}) = 0.055926.$$

Endpoints do not affect probability for a continuous variable.

Q7(b). LED lifespan: critical-value t-test

Question. Claimed mean bulb life is 8000 hours. Lifespans of 12 bulbs, in hundreds of hours: 78, 85, 75, 82, 79, 88, 80, 81, 74, 86, 83, 76. Test the claim at 1%.

Worked solution. Keep the units consistent: 8,000 hours corresponds to the null mean 80 in hundreds of hours.Use a one-sample t-test because the population standard deviation is unknown. Assume independent random observations from an approximately normal population. Let μ be the true mean, $\mu_0 = 80$ the claimed mean, $\bar{x}$ the sample mean, s the sample standard deviation, n the sample size, and $\nu = n - 1$ the degrees of freedom. Test $H_0 : \mu = 80$ against $H_1 : \mu \neq 80$.

$$n = 12, \qquad \bar{x} = 80.58333, \qquad \sum_{i=1}^{n}(x_i - \bar{x})^2 = 216.91667.$$

Here x_i is observation i. Therefore

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}} = 4.44069, \qquad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = 0.45505.$$

At significance level $\alpha = 0.01$, $\nu = 11$ and the two-sided critical values are $\pm$ 3.10581. Do not reject H_0, since $|t|$ is smaller than the positive critical value. The observations are compatible with the claimed mean; this does not prove that it is exactly correct.

8 A compact examination checklist

1. Define the variable and its units before choosing a model. Distinguish a count from the waiting time to an event.
2. For grouped data, write frequencies, midpoints, and cumulative frequencies first. Convert cumulative counts to class counts by subtraction.
3. Choose the variance divisor deliberately: N describes the listed distribution; $n-1$ estimates a population variance from a sample. Here N is a distribution's total count and n a sample size.
4. For strict count inequalities, list the allowed integers before summing. "Less than 7" ends at 6.
5. For conditional probability, identify the event after the bar. That event determines the denominator.
6. For a transformed discrete variable, combine probabilities from all inputs producing the same output.
7. For a hypothesis test, state the hypotheses, assumptions, significance level, statistic, degrees of freedom, critical region, decision and contextual conclusion. A two-sided t-test divides the significance level between both tails.
8. State missing assumptions. A mean and standard deviation alone do not specify a distribution, and separate variances do not specify covariance.
9. Round after the calculation, and do not interpret failure to reject as proof of equality.

A Original examination papers

The following pages reproduce all eight supplied files as resized viewing copies. The unchanged originals are retained alongside them in the sources folder. The appendix is a source record; worked solutions appear in the preceding year-wise chapters. P2 and P8 are identical, and are reproduced separately to preserve the full set of uploads.

P1

`Managing_Uncertainty_2022-23___Final_GGB9VPY3a7.pdf`

P2

`Managing_Uncertainty_2025-26_Final_xWtQQHWW7F.pdf`

P3

`Managing_Uncertainty___Final__Re-exam_2022-23_CnPzWPv5Aw.pdf`

P4

`Managing_Uncertainty___Re-exam_(_2024-25)_(_2022-23_2023-24)_kloI34FiYa.pdf`

P5

`Managing_Uncertainty___Re-exam_2022-23_LD1iyiVsPU.pdf`

P6

`Managing_Uncertainty_Batch_2024-25_Final_Batch_2022-23_2023-24_Re_Exam_2HDhT9krDq.pdf`

P7

`Managing_Uncertainty_DmYBLzOFv0.pdf`

P8

`Managing_Uncertainty_mSBF5Fjshq(1).pdf`

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Academic Year: 2022-23

Programme: B.Tech /MBA Tech (Data Science) Year: II Semester: III

Subject: Managing Uncertainty

Date: 21 November 2022

Marks: 100
Time: 10.00 am - 1.00 pm
Durations: 3 (Hrs)
No. of Pages: 05+1=6

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	20
CO- 2; SO-1 ; BL-1	A	Suppose that samples of polythene bags from two manufacturers A and B are tested by a buyer for bursting pressure, giving the following results	

	Number of Bags	
Bursting Pressure	A	B
5 -10	2	9
10 -15	9	11
15 – 20	29	18
20 – 25	54	32
25 – 30	11	27
30 – 35	5	13

2

3

1/6

<table>
<tr><td></td><td></td><td>I. Which set of bags has the on an average highest bursting pressure?

II. Which has more uniform pressure? If prices are the same, which manufacture's bag would be preferred by the buyer? Why?</td><td></td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-2</td><td>B</td><td>In rolling two dice, what is the probability of rolling

I. A total of 7 on the first roll, followed by a total of 11 on the second roll?

II. A total of 21 on the first two rolls combined?

III. A total of 6 on the first three rolls combined?</td><td>2
2
1</td></tr>
<tr><td>CO-1 ;
SO-1 ;
BL-2</td><td>C</td><td>Briefly describe Frequency Polygons.</td><td>5</td></tr>
<tr><td>CO- 3;
SO-2 ;
BL-1</td><td>D</td><td>In an attempt to estimate potential future demand, the National Motor Company did a study asking married couples how many cars the average energy-minded family should own in 1998. For each couple, National averaged the husband's and wife's response to get the overall couple response. The answers were then tabulated.
<table><tr><td>Number of cars</td><td>0</td><td>0.5</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td></tr><tr><td>Frequency</td><td>2</td><td>14</td><td>23</td><td>7</td><td>4</td><td>2</td></tr></table>Calculate the variance and the standard deviation.</td><td>5</td></tr>
<tr><td>Q2</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q2
CO-2;
SO-1;
BL-1</td><td>A</td><td>Hinton Press hypothesizes that the average life of its largest web press is 14,500 hours. They know that the standard deviation of press life is 2,100 hours. From a sample of 25 presses, the company finds a sample mean of 13,000 hours. At a 0.01 significance level, should the company conclude that the average life of the presses is less than the hypothesized 14,500 hours?</td><td>10</td></tr>
</table>

2/6/

<table>
<tr><td>Q2
CO-2;
SO-1;
BL-2</td><td>B</td><td>A study of the age of 100 persons grouped into intervals 20-22, 22-24, 24-26,…., revealed the mean age and standard deviation to be 32.02 and 13.18 respectively. While checking it was discovered that the observation 57 was misread as 27. Calculate the correct mean age and standard deviation.</td><td>10</td></tr>
<tr><td>Q3</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q3
CO-3;
SO-2; BL-3</td><td>A</td><td>Calculate the standard deviation and coefficient of variation from the following data.
<table><tr><th>Marks</th><th>No of students</th><th>Marks</th><th>No of students</th></tr><tr><td>Up to 10</td><td>12</td><td>Up to 50</td><td>157</td></tr><tr><td>Up to 20</td><td>30</td><td>Up to 60</td><td>202</td></tr><tr><td>Up to 30</td><td>65</td><td>Up to 70</td><td>222</td></tr><tr><td>Up to 40</td><td>107</td><td>Up to 80</td><td>230</td></tr></table></td><td>10</td></tr>
<tr><td>Q3
CO-2;
SO-1; BL-2</td><td>B</td><td>Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112. What is the probability of an individual scoring above 500 on the GMAT? How high must an individual score on the GMAT in order to score in the highest 5%?</td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q4
CO-;2
SO-2;
BL-1</td><td>A</td><td>A company wants to improve sales. Past sales data indicate that the average sale was $100 per transaction. After training the sales force, recent sales data (taken from a sample of 25 salesmen) indicates an average sale of $130, with a standard deviation of $15. Did the training work? Test your hypothesis at a 5% significance level.</td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be
I. 0 jobs;</td><td>10</td></tr>
</table>

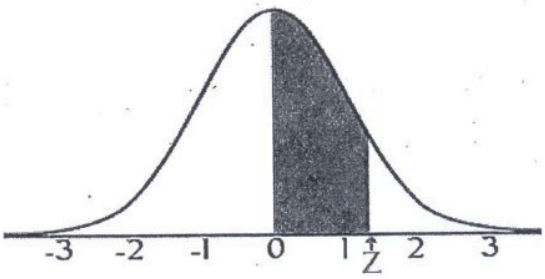
3/6/

CO-;2 SO-2; BL-1	B	II. exactly 2 jobs; III. at most 3 arrivals. IV. What is the maximum jobs that should arrive one minute with 90 % certainty? V. more than 3 Jobs	
Q5		Answer briefly:	20
Q5 CO-2; SO-2; BL-3	A	Two dice are rolled. A = 'sum of two dice equals 3' B = 'sum of two dice equals 7' C = 'at least one of the dice shows a 1' I. What is P(A|C)? II. What is P(B|C)? III. Are A and C independent? What about B and C?	3 3 4
Q5 CO-2; SO-2; BL-2	B	In its third year, attendance in the Liberty Football League averaged 16,050 fans per game,and had a standard deviation of 2,500. I. According to these data, what is the probability that the number of fans at any given game was greater than 20,000? II. Fewer than 10,000? III. Between 14,000 and 17,500?	10
Q6		Answer briefly:	20
Q6 CO-2; SO-2; BL-3	A	Hunter says she is itchy. There is a test for Allergy to Cats, but this test is not always right: • For people that really do have the allergy, the test says "Yes" 80% of the time • For people that do not have the allergy, the test says "Yes" 10% of the time ("false positive") If 1% of the population have the allergy, and Hunter's test says "Yes", what are the chances that Hunter really has the allergy?	10
Q6		You are planning a picnic today, but the morning is cloudy • Oh no! 50% of all rainy days start off cloudy!	10

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CO-2; SO-2; BL-2	B	• But cloudy mornings are common (about 40% of days start cloudy) • And this is usually a dry month (only 3 of 30 days tend to be rainy, or 10%) What is the chance of rain during the day?	
Q7		Answer briefly:	20
Q7 CO-2; SO-2; BL-2	A	I. Explain continuous random variables. Also, give and explain their PDF. II. State and explain joint probability distribution for continuous random variables.	6 4
Q7 CO-2; SO-2; BL-1	B	In the past 20 years, on average, only 3 percent of all checks written to the American Heart Association have bounced. This month, the A.H.A. received 200 checks. What is the prob- ability that I. Exactly 10 of these checks bounced? II. Exactly 5 of these checks bounced?	10

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STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

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Programme: B. Tech (DS)/B. Tech Intg. (DS)/MBA Tech (DS)
Year: II / IV/Semester III/Semester VII(Exam Year: 2025-2026)
Subject: Managing Uncertainty
Date: 29 Nov 2025 **Time: 10:00 am to 01:00 pm (03:00 Hrs.)**
Max. Marks: 100

FINAL EXAMINATION (Acad. Year:**2025-2026**)

Instructions:

1. This question paper contains 4 pages.
2. Question No. 1 is compulsory.
3. Out of remaining questions, attempt any 4 questions.
4. In all 5 questions to be attempted.
5. All questions carry equal marks.
6. Answer to each new question to be started on a fresh page.
7. Figure in brackets on the right-hand side indicates full marks
8. Assume Suitable data if necessary.

<table>
<tr><td>Q1</td><td>Q.1</td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ; BL-3</td><td>a.</td><td>The following table gives the frequency distribution of the weekly wages (in '00 Rs.) of 100 workers in a factory.
<table>
<tr><td>Weekly Wages (₹00)</td><td>20–24</td><td>25–29</td><td>30–34</td><td>35–39</td><td>40–44</td><td>45–49</td><td>50–54</td><td>55–59</td><td>60–64</td></tr>
<tr><td>Number of Workers</td><td>4</td><td>5</td><td>12</td><td>23</td><td>31</td><td>10</td><td>8</td><td>5</td><td>2</td></tr>
</table>
Develop the histogram and frequency polygon of the distribution.</td><td>[5]</td></tr>
<tr><td>CO-1,5 ; BL-2</td><td>b.</td><td>Compare Binomial and Poisson Distribution</td><td>[5]</td></tr>
<tr><td>CO-2 ; BL-3</td><td>c.</td><td>A bag contains 7 white and 9 black balls. Two balls are drawn in succession without replacement at random. Identify the probability that one of them is white and the other is black</td><td>[5]</td></tr>
<tr><td>CO-1 ; BL-2</td><td>d.</td><td>Explain the Qualitative and Quantitative types of classification in detail</td><td>[5]</td></tr>
</table>

********** 1 **********

Q2.a CO-3,4 ; BL-4		Distinguish Karl Pearson's coefficient of skewness from the following data and explain its significance Wages (₹): 70–80 | 80–90 | 90–100 | 100–110 | 110–120 | 120–130 | 130–140 | 140–150 No. of Persons: 12 | 18 | 35 | 42 | 50 | 45 | 20 | 8	[10]
Q2.b CO-2 ; BL-4		Simplify and State Bayes' theorem and provide a detailed proof along with an explanation of its components and applications.	[10]
Q3 CO-3 ; BL-5		In the following data, two class frequencies are missing. Class Interval: 100-110 | 110-120 | 120-130 | 130-140 | 140-150 | 150-160 | 160-170 | 170-180 | 180-190 | 190-200 Frequency: 4 | 7 | 15 | f1 | 40 | f2 | 16 | 10 | 6 | 3 However, it was possible to ascertain that the total number of frequencies was 150 and that the median has been correctly found out as 146·25. You are required to Evaluate with the help of information given : (i) The two missing frequencies. (ii) Having found the missing frequencies, calculate arithmetic mean and standard deviation. (iii) Find the value of mode.	[20]
Q4.a CO-2,6 ; BL-5		The joint probability function of X and Y is given by $p(x,y)=k(x^2+y)$ for x=1,2 and y=1,2,3. Determine: (i) The value of k. (ii) The marginal distributions of X and Y. (iii) Are X and Y independent? (iv) The conditional distribution of X for Y=2.	[10]
Q4.b CO-2 ; BL-3		Ten unbiased coins are tossed simultaneously. Identify the probability of obtaining; (i) Exactly 6 heads (ii) At least 8 heads (iii) No head (iv) At least one head (v) Not more than three heads (vi) At least 4 heads.	[10]
Q5.a CO-2,5 ; BL-3		The time (in minutes), a customer spends waiting for a service agent at a call center follows an Exponential Distribution with a mean waiting time of 5 minutes. (a) Determine the probability density function (PDF) and the cumulative distribution function (CDF) for the waiting time, X. (b) Find the probability that a customer waits for more than 8 minutes. (c) Find the probability that a customer waits between 3 and 7 minutes.	[10]

<table>
<tr><td>Q5.b
CO-2 ;
BL-4</td><td></td><td>There are three identical-looking boxes: B1, B2, and B3. The composition of the balls in each box is as follows:
Box B1: 5 Red balls, 5 Green balls
Box B2: 2 Red balls, 8 Green balls
Box B3: 7 Red balls, 3 Green balls

A box is chosen at random, and then a ball is drawn from that box. The drawn ball is observed to be Red. Analyze is the probability that the ball came from Box B3?</td><td>[10]</td></tr>
<tr><td>Q6.a
CO-3,6 ;
BL-2</td><td></td><td>Explain Type I and Type II error. Explain Skewness and Kurtosis in detail</td><td>[10]</td></tr>
<tr><td>Q6.b
CO-3 ;
BL-3</td><td></td><td>The number of critical errors detected per hour on a new e-commerce website was observed over a 200-hour testing period.

<table><tr><td>Number of Errors per Hour (X)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Observed Frequency (f)</td><td>120</td><td>55</td><td>18</td><td>5</td><td>2</td></tr></table>
Identify the expected frequencies using the Poisson distribution. Also, compute the mean and variance of the fitted (theoretical) distribution.</td><td>[10]</td></tr>
<tr><td>Q7.a
CO-2;
BL-5</td><td></td><td>A continuous random variable X has the following probability density function
$$f(x) = \frac{a}{x^5}, 2 \leq X \leq 10$$
Determine the constant a, distribution function of X and find the probability of the event $4 \leq X \leq 7$</td><td>[10]</td></tr>
<tr><td>Q7.b
CO-5,6 ;
BL-6</td><td></td><td>A manufacturer claims that the mean lifespan of their new LED bulb is 8,000 hours. A consumer testing agency randomly selected 12 bulbs and recorded their actual lifespans (in hundreds of hours):

78,85,75,82,79,88,80,81,74,86,83,76

Test whether these results support the manufacturer's claim? Use a significance level of α=0.01.</td><td>[10]</td></tr>
</table>

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2023-2024

Program/s: MBATech/BTech

Year: II Semester: III

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 03 hrs (10 am to 1 pm)

Date: 25 / 11 / 2023

No. of Pages: 05

Marks: 100

Final Examination/Re-Examination (2022-2023)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

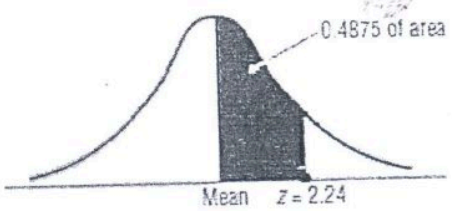
1) Question No. 1 is compulsory.
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7) **Assume Suitable data if necessary.**

Q1 CO- 1; SO- 1; BL-2	a.	Explain classification. Also, enlist various types of classification.	[5]
CO- 1; SO-1 ; BL-1	b.	Explain the scatter plot with a suitable example.	[5]
CO-2 ; SO-2 ; BL-3	c.	A recent study of how IT professionals spend their leisure time surveyed workers employed for more than 5 years. They determined the probability of an employee having 2 weeks of vacation time to be 0.45, 1 week of vacation time to be 0.10, and 3 or more weeks of vacation time to be 0.20. Suppose 20 workers are selected at random. Answer the following questions. a) What is the probability that 8 have 2 weeks of vacation time? b) What is the probability that at most 2 of the workers have 3 or more weeks of vacation time?	[5]

<table>
<tr><td>CO- 3;
SO- 2;
BL-2</td><td>d.</td><td>For the given observations for random variables X and Y, calculate values for the following transformations.
E(X)=10, V(X)=3, E(Y)=20, V(Y)=2
a) E(3Y-7)
b) E(X-Y)
c) V(X-Y)
d) V(5X+9)</td><td>[5]</td></tr>
<tr><td>Q2
CO-1;
SO-2;
BL-5</td><td>a.</td><td>In Tarun's school, there are 25 teachers. Each teacher travels to school every morning in his or her own car. The distribution of the driving times (in minutes) from home to school for the teachers is shown in the table below:
Driving time (in minutes) Number of Teachers
0-10 3
10-20 10
20-30 6
30-40 4
40-50 2
Calculate the mean of the driving times.</td><td>[10]</td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-5</td><td>b.</td><td>Calculate the correlation coefficient of the given data
<table><tr><td>x</td><td>12</td><td>15</td><td>18</td><td>21</td><td>27</td></tr><tr><td>y</td><td>2</td><td>4</td><td>6</td><td>8</td><td>12</td></tr></table>Comment and explain the type of relationship that variables x and y exhibit from your answers.</td><td>[10]</td></tr>
<tr><td>Q3
CO-2;
SO-2;
BL-3</td><td>a.</td><td>The following data of the daily production rate of fibreglass boats from Hydrosport, Ltd. a Miami manufacturer,
17 21 18 27 17 21 20 22 18 23
The company production manager feels that a standard deviation of more than three boats a day indicates unacceptable production-rate variations. Should she be concerned about plant production rates?</td><td>[10]</td></tr>
</table>

CO-2 ; SO- 6; BL-3	b.	A bag contains 3 red and 7 black balls. Two balls are drawn at random without replacement. If the second ball is red, what is the probability that the first ball is also red?	[10]
Q4 CO-2; SO-6; BL-2	a.	Three urns are there containing white and black balls; the first urn has 3 white and 2 black balls; the second urn has 2 white and 3 black balls and the third urn has 4 white and 1 black balls. Without any biasing, one urn was chosen, and from that one ball was chosen randomly which was white. What is the probability that it came from the third urn?	[5]
CO-3; SO-6; BL-3	b.	On average, a pair of running shoes can last 18 months. The length of time running shoes last is exponentially distributed. What is the probability that a pair of running shoes last more than 15 months?	[5]
CO-3; SO-2; BL-3	c.	The average weight gained by a person over the winter months is uniformly distributed and ranges from 0 to 30 lbs. Find the probability that a person will gain between 10 and 15 lbs in the winter months. Also, calculate the expected value and standard deviation for the given distribution.	[10]
Q5 CO-2; SO-2; BL-2	a.	Calculate the variance for the given data x_i f_i 10 1 4 3 6 5 8 1	[10]
CO- 3; SO- 6; BL-5	b.	If 3% of electronic units manufactured by a company are defective. Find the probability that in a sample of 200 units, less than 2 bulbs are defective.	[10]
Q6 CO-3; SO-2; BL-3	a.	How the range of a function of a random variable X can be generated using the range of X. Let X be a discrete random variable with $P_X(k)=1/4$ for $k=-1,0,1,2$. Let $Y=3\|X\|$. Find the range and PMF of Y.	[10]
CO-3 ; SO- 1; BL-1	b.	Describe the concept of PMF for a discrete random variable with a suitable example. Also, give PDF for exponential distribution.	[10]

Q7 CO-3; SO-6; BL-5	a.	The average number of acres burned by forest and range fires in a large New Mexico county is 4,300 acres per year, with a standard deviation of 750 acres. The distribution of the number of acres burned is normal. What is the probability that between 2,500 and 4,200 acres will be burned in any given year?	[10]
CO- 3; SO- 6; BL-5	b.	In 1993, the Financial Accounting Standards Board (FASB) was considering a proposal to require companies to report the potential effect of employees' stock options on earnings per share (EPS). A random sample of 41 high-technology firms revealed that the new proposal would reduce EPS by an average of 13.8 percent, with a standard deviation of 18.9 percent. A random sample of 35 producers of consumer goods showed that the proposal would reduce EPS by 9.1 percent on average, with a standard deviation of 8.7 percent. On the basis of these samples, is it reasonable to conclude (at $\alpha = 0.10$) that the FASB proposal will cause a greater reduction in EPS for high-technology firms than for producers of consumer goods?	[10]



EXAMPLE: TO FIND THE AREA UNDER THE CURVE BETWEEN THE MEAN AND A POINT 2.24 STANDARD DEVIATIONS TO THE RIGHT OF THE MEAN, LOOK UP THE VALUE OPPOSITE 2.2 AND UNDER 0.04 IN THE TABLE; 0.4875 OF THE AREA UNDER THE CURVE LIES BETWEEN THE MEAN AND A z VALUE OF 2.24.

APPENDIX TABLE 1 AREAS UNDER THE STANDARD NORMAL PROBABILITY DISTRIBUTION BETWEEN THE MEAN AND POSITIVE VALUES OF z

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech /MBA Tech/BTI

Year: II/IV Semester: III/VII

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 3 hrs (10.00 a.m to 1.00 p.m)

Date: 22/2/25

No. of Pages:5

Marks: 100

Re Examination (2024-25) /Re-Examination (2022-23, 23-24)

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<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-2</td><td>a.</td><td>Explain the objective of Tabulation. Explain the difference between classification and tabulation</td><td>[5]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-3</td><td>b.</td><td>The data below give the yearly profits (in thousands of rupees) of two companies A and B. Develop the data by means of a suitable bar diagram.
<table>
<tr><td>Year</td><td>Company A</td><td>Company B</td></tr>
<tr><td>1994-95</td><td>120</td><td>90</td></tr>
<tr><td>1995-96</td><td>135</td><td>95</td></tr>
<tr><td>1996-97</td><td>140</td><td>108</td></tr>
<tr><td>1997-98</td><td>160</td><td>120</td></tr>
<tr><td>1998-99</td><td>175</td><td>130</td></tr>
</table></td><td>[5]</td></tr>
</table>

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<table>
<tr><td>CO-1 ;
SO-2 ;
BL-1</td><td>c.</td><td>An urn contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that (i) both are white, (ii) both are red, (iii) one is of each color.</td><td>[5]</td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-2</td><td>d.</td><td>Explain Type I and Type II error</td><td>[5]</td></tr>
<tr><td>Q2.a
CO-3;
SO-2;
BL-5</td><td></td><td>Determine Karl-Pearson's coefficient of skewness for the given distribution:
<table>
<tr><td>X:</td><td>0-5</td><td>5-10</td><td>10-15</td><td>15-20</td><td>20-25</td><td>25-30</td><td>30-35</td><td>35-40</td></tr>
<tr><td>F:</td><td>2</td><td>5</td><td>7</td><td>13</td><td>21</td><td>15</td><td>8</td><td>3</td></tr>
</table></td><td>[10]</td></tr>
<tr><td>Q2.b
CO-3;
SO-2;
BL-5</td><td></td><td>For the following frequency distribution, determine (a) The median class. (b) The number of the item that represents the median. (c) The width of the equal steps in the median class. (d) The estimated value of the median for these data. (e) Estimate the value of Median with the formula.
<table>
<tr><th>Class in Dollars</th><th>Frequency</th></tr>
<tr><td>0–49.99</td><td>78</td></tr>
<tr><td>50.00– 99.99</td><td>123</td></tr>
<tr><td>100.00–149.99</td><td>187</td></tr>
<tr><td>150.00–199.99</td><td>82</td></tr>
<tr><td>200.00–249.99</td><td>51</td></tr>
<tr><td>250.00–299.99</td><td>47</td></tr>
<tr><td>300.00–349.99</td><td>13</td></tr>
<tr><td>350.00–399.99</td><td>9</td></tr>
<tr><td>400.00–449.99</td><td>6</td></tr>
<tr><td>450.00–499.99</td><td>2</td></tr>
<tr><td></td><td>600</td></tr>
</table></td><td>[10]</td></tr>
<tr><td>Q3.a
CO-1;
SO-2;
BL-6</td><td></td><td>We have data from 100 economists in academics, private sector, and government, concerning their opinions whether the economy would be stable, expand or contract in the near future. However, a part of the information was lost. Based on remaining data, you are required to create a probability table.
From the given table find : (i) P(A) ; (ii) P(E) ; (iii) P(A ∩ E) ;
(iv) P(G ∩ C) ; (v) P (E | G) ; (vi) P (G | E).</td><td>[10]</td></tr>
</table>

		<table><tr><th rowspan="2">Economists</th><th colspan="3">Economy</th><th rowspan="2">Total</th></tr><tr><th>Stable (S)</th><th>Expanding (E)</th><th>Contracting (C)</th></tr><tr><td>Academics (A)</td><td>25</td><td></td><td>20</td><td></td></tr><tr><td>Private Sector (R)</td><td></td><td>7</td><td></td><td>22</td></tr><tr><td>Government (G)</td><td>5</td><td>8</td><td></td><td>13</td></tr><tr><td></td><td>40</td><td></td><td></td><td></td></tr></table>	
Q3.b CO-1; SO-2; BL-5		The results of an investigation by an expert on a fire accident in a skyscraper are summarized below : (i) Prob. (there could have been short circuit) = 0·8 (ii) Prob. (LPG cylinder explosion) = 0·2 (iii) Chance of fire accident is 30% given a short circuit and 95% given an LPG explosion. Based on these, what do you think is the most probable cause of fire? Statistically justify your answer.	[10]
Q4.a CO-2; SO-2; BL-5		The random variable X has binomial distribution B 15, 0.35 ( ). Determine each of the following. a) P (X = 8). b) P (X < 7). c) P (X > 9). d) P (5< X< 10).	[10]
Q4.b CO-2; SO-2; BL-5		A continuous function of a random variable X has the p.d.f $f(x) = kx^2e^{-x}$, x>=0. Evaluate k, mean and variance.	[10]
Q5.a CO-2; SO-2; BL-3		It is given that 2% of the test tubes are defective. Apply Poisson Distribution to find the probability that a packet of 100 test tubes contains: (1) No defective test tubes (2) One defective test tubes (3) Two or more defective test tubes (4) Two or less defective test tubes.	[10]
Q5.b CO-2; SO-2; BL-		Suppose that in an Olympics game Archer shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7. Determine the (i) Probability that the target would hit on the 10th attempt?(ii) probability that it taken him less than 4 shots? (iii) Probability that it taken him an even number of shots? (iv) Average number of shots needed to hit the target.	[10]
Q6.a		The joint density function of A and B is given by P(a, b) = k(2a+b); a = 0,1,2; b = 0,1,2	[10]

<table>
<tr><td>CO-1;
SO-2;
BL-4</td><td></td><td>Distinguish (i) value of k (ii) Marginal distribution of A and B (iii) are A and B independent? (iv) find the conditional distribution of B for A = a.</td><td></td></tr>
<tr><td>Q6.b
CO-3;
SO-2;
BL-4</td><td></td><td>In a population, the average IQ is 100. A team of scientists wants to test a new medication to see if it has either a positive or negative effect on intelligence or no effect at all. A sample of 30 participants who has taken the medication has a mean of 140 with a standard deviation of 20. Did the medication affect intelligence? Alpha = 0.05</td><td>[10]</td></tr>
<tr><td>Q7.a
CO-3;
SO-2;
BL-2</td><td></td><td>State Bayes theorem. Explain in detail with proof</td><td>[10]</td></tr>
<tr><td>Q7.b
CO-1;
SO-2;
BL-4</td><td></td><td>Two independent random samples of size 4 and 5 are selected from each of the normal populations
<table>
<tr><td>Population 1</td><td>12</td><td>3</td><td>8</td><td>5</td><td></td></tr>
<tr><td>Population 2</td><td>14</td><td>7</td><td>7</td><td>9</td><td>6</td></tr>
</table>
(1) calculate S2 the pooled estimator of variance
(2) find the 95% confidence interval for $\mu 1 - \mu 2$, the difference between the two population
(3) Test H0: $\mu 1 - \mu 2 = 0$ against H1: $\mu 1 - \mu 2 \neq 0$ for alpha = 0.05. state your conclusion</td><td>[10]</td></tr>
</table>

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t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

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Academic Year: 2023-2024

Program/s: MBATech/BTech

Year: II Semester: III

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 03 hrs (10:00am to 1:00pm)

Date: 08 /02 / 2024

No. of Pages: 05

Marks: 100

Re-Examination (2022-23)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

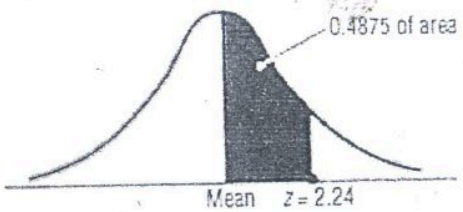
1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1 CO- 1; SO- 1; BL-2	a.	Explain differences between classification and tabulation.	[5]
CO- 1; SO-1 ; BL-1	b.	Explain dot plots with suitable example.	[5]
CO-2 ; SO-2 ; BL-3	c.	On average, every one out of 10 telephones is found busy. Six telephone numbers are selected at random. Find the probability that four of them will be busy.	[5]
CO- 3; SO- 2; BL-2	d.	For the given observations for random variables X and Y, calculate values for the following transformations. E(X)=15, V(X)=4, E(Y)=30, V(Y)=2 a) E(4Y-8) b) E(X-Y) c) V(X-Y) d) V(4X+8)	[5]

<table>
<tr><td>Q2
CO-1;
SO-2;
BL-5</td><td>a.</td><td>From the following data:
<table><tr><th>Class</th><th>Frequency</th></tr><tr><td>20-29</td><td>6</td></tr><tr><td>30-39</td><td>16</td></tr><tr><td>40-49</td><td>21</td></tr><tr><td>50-59</td><td>29</td></tr><tr><td>60-69</td><td>25</td></tr><tr><td>70-79</td><td>22</td></tr><tr><td>80-89</td><td>11</td></tr><tr><td>90-99</td><td>7</td></tr><tr><td>100-109</td><td>4</td></tr><tr><td>110-119</td><td>0</td></tr><tr><td>120-129</td><td>2</td></tr></table>Compute Sample Mean using the Step-deviation/Coding method with 0 assigned to the 60-69 class.
Instruction – Do not make classes continuous.</td><td>[10]</td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-5</td><td>b.</td><td>Calculate Spearman's rank correlation coefficient from the following data:
x 10 12 8 15 20 25 40
y 15 10 6 25 16 12 8
Comment and explain the type of relationship that variables x and y exhibit from your answers.</td><td>[10]</td></tr>
<tr><td>Q3
CO-2;
SO-2;
BL-3</td><td>a.</td><td>The number of checks cashed each day at five branches of the Bank of Orange County during the past month had the following frequency distribution:
<table><tr><th>Class</th><th>Frequency</th></tr><tr><td>0-199</td><td>10</td></tr><tr><td>200-399</td><td>13</td></tr><tr><td>400-599</td><td>17</td></tr><tr><td>600-799</td><td>42</td></tr><tr><td>800-999</td><td>18</td></tr></table>Hank Spivey, director of operations for the bank, knows that a standard deviation in check cashing of more than 200 checks per day creates staffing and organizational problems at the branches because of uneven workload. Should Hank worry about staffing next month?</td><td>[10]</td></tr>
</table>

CO-2 ; SO- 6; BL-3	b.	A box contains three coins: two regular coins and one fake two-headed coin (P(H)=1), a) You pick a coin at random and toss it. What is the probability that it lands heads up? b) You pick a coin at random and toss it, and get heads. What is the probability that it is the two-headed coin?	[10]
Q4 CO-2; SO-6; BL-2	a.	Three identical boxes contain red and white balls. The first box contains 3 red and 2 white balls, the second box has 4 red and 5 white balls, and the third box has 2 red and 4 white balls. A box is chosen very randomly and a ball is drawn from it. If the ball that is drawn out is red, what will be the probability that the second box is chosen?	[5]
CO-3; SO-6; BL-3	b.	The total duration of baseball games in the major league in the 2011 season is uniformly distributed between 447 hours and 521 hours inclusive. • Find the mean. • What is the probability that the duration of games for a team for the 2011 season is between 480 and 500 hours?	[5]
CO-3; SO-2; BL-3	c.	Assume that you usually get 2 phone calls per hour. calculate the probability, that a phone call will come within the next hour. Also, calculate the expected value and standard deviation for the given distribution.	[10]
Q5 CO-2; SO-2; BL-2	a.	Find the variance of the following data table **Class** **Frequency** 0-10 3 10-20 6 20-30 4 30-40 2 40-50 1	[10]
CO- 3; SO- 6; BL-5	b.	In Lucknow Railway Station, recently there was a complaint about the accidents on its elevators and the resulting injuries, especially among the elderly people. According to past records, on average, five people met with	[10]

		an accident every week. The station in charge has requested the finance department to allocate extra funds for purchasing safety devices to check the accidents on the elevators. The finance department has replied that funds cannot be allocated unless the chances of more than three accidents on elevators in any week exceed 70 percent. Will funds be allocated?	
Q6 CO-3; SO-2; BL-3	a.	How PMF of a function of a random variable X can be generated using the PMF of X. Let X be a discrete random variable with PX(k)=1/4 for k=−1,0,1,2. Let $Y=X^2$. Find the range and PMF of Y.	[10]
CO-3 ; SO- 1; BL-1	b.	Describe the concept of PDF and CDF for a continuous random variable with a suitable example.	[10]
Q7 CO-3; SO-6; BL-5	a.	The length of human pregnancies from conception to birth approximates a normal distribution with a mean of 266 days and a standard deviation of 16 days. What proportion of all pregnancies will last between 240 and 270 days (roughly between 8 and 9 months)?	[10]
CO- 3; SO- 6; BL-5	b.	Two independent samples of observations were collected. For the first sample of 60 elements, the mean was 86 and the standard deviation 6. The second sample of 75 elements had a mean of 82 and a standard deviation of 9. (a) Compute the estimated standard error of the difference between the two means. (b) Using $\alpha = 0.01$, test whether the two samples can reasonably be considered to have come from populations with the same mean.	[10]



EXAMPLE: TO FIND THE AREA UNDER THE CURVE BETWEEN THE MEAN AND A POINT 2.24 STANDARD DEVIATIONS TO THE RIGHT OF THE MEAN, LOOK UP THE VALUE OPPOSITE 2.2 AND UNDER 0.04 IN THE TABLE; 0.4875 OF THE AREA UNDER THE CURVE LIES BETWEEN THE MEAN AND A z VALUE OF 2.24.

APPENDIX TABLE 1 AREAS UNDER THE STANDARD NORMAL PROBABILITY DISTRIBUTION BETWEEN THE MEAN AND POSITIVE VALUES OF z

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech /MBA Tech/BTI | Year: II/IV Semester: III/VII

Stream/s : Data Science

Subject: Managing Uncertainty | Time: 3 hrs (10.00 a.m to 1.00 p.m)

Date: 11/12/24 | No. of Pages:5

Marks: 100

Final Examination (2024-25) /Re-Examination (2022-23, 23-24)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-2</td><td>a.</td><td>Explain different types of classification.</td><td>[5]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-3</td><td>b.</td><td>The following data relating to the strength of the Indian Merchant Shipping Fleet gives the Gross Registered Tonnage (GRT) as on 31st December, for different years.
<table><tr><td>Year</td><td>1961</td><td>1966</td><td>1971</td><td>1975</td><td>1976</td></tr><tr><td>GRT in '000</td><td>901</td><td>1792</td><td>2500</td><td>4464</td><td>5115</td></tr></table>
Construct the data by suitable bar diagram.</td><td>[5]</td></tr>
<tr><td>CO-1 ;
SO-2 ;
BL-1</td><td>c.</td><td>In a single throw with two uniform dice find the probability of throwing (i) Five, (ii) Eight.</td><td>[5]</td></tr>
</table>

1/5

CO-2 ; SO-2 ; BL-2	d.	Explain PMF, PDF and CDF. Also, explain the relation between them.	[5]
Q2.a CO-3; SO-2; BL-5		The following are the marks of 150 students in an examination. Evaluate Karl Pearson's coefficient of skewness. Comment on the answer	[10]

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students	10	40	20	0	10	40	16	14

Q2.b CO-3; SO-2; BL-5

For the following frequency distribution, determine (a) The median class. (b) The number of the item that represents the median. (c) The width of the equal steps in the median class. (d) The estimated value of the median for these data. (e) Estimate the value of Median with the formula. [10]

Class	Frequency
100–149.5	12
150–199.5	14
200–249.5	27
250–299.5	58

Class	Frequency
300–349.5	72
350–399.5	63
400–449.5	36
450–499.5	18

Q3.a CO-1; SO-2; BL-5

A part is observed to have no defect. Estimate the probability that it was produced by X manufacturer? (ii) A part is known to have been produced by manufacturer X. Estimate the probability that the part has no defects? (iii) A part is known to have two or more defects. Estimate the probability that it was manufactured by X? (iv) A part is known to have one or more defects. Estimate the probability that it was manufactured by Y? [10]

Manufacturer	Number of Defects			
	0	1	2	3
X	0.1250	0.0625	0.1875	0.1250
Y	0.0625	0.0625	0.1250	0.2500

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Q3.b CO-1; SO-2; BL-5		A company launches an advertising campaign of its new product on TV, radio and in print media in an area where 30% watch TV, 50% listen to the radio and the rest rely on newspapers for all information. It is estimated that a person who sees the advertisement on TV will buy the product with probability 0·6. A person who has heard it on radio is expected to buy the product with probability 0·3 and seeing the advertisement in print will convince a person to buy the product with probability 0·1. A consumer, chosen at random, is found to have purchased the product. Evaluate the probability she heard about the product on radio?	[10]
Q4.a CO-2; SO-2; BL-5		The random variable X has binomial distribution B (20, 0.4). Determine each of the following. a) P (X = 6). b) P (X ≤ 9). c) P (X ≥ 12). d) P (8 ≤ X≤ 13).	[10]
Q4.b CO-2; SO-2; BL-5		A discrete random variable has the probability density function given below. Evaluate k, mean and variance X: -2, -1, 0, 1, 2, 3 P(X=x): 0.2, k, 0.1, 2k, 0.1, 2k	[10]
Q5.a CO-2; SO-2; BL-4		It is given that 2% of the screws are defective. Apply Poisson Distribution to determine the probability that a packet of 100 screws contains: (1) No defective screws (2) One defective screw (3) Two or more defective screw (4) Two or less defective screw	[10]
Q5.b CO-2; SO-2; BL-4		Suppose that a trainee soldier shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7. Determine the (i) Probability that the target would hit on the 10th attempt? (ii) probability that it taken him less than 4 shots? (iii) Probability that it taken him an even number of shots? (iv) Average number of shots needed to hit the target.	[10]
Q6.a CO-1; SO-2; BL-4		The joint density function of X and Y is given by P(x,y) = k(2x+y); x = 0,1,2; y = 0,1,2 Distinguish (i) value of k (ii) Marginal distribution of X and Y (iii) are X and Y independent? (iv) find the conditional distribution of Y for X = x.	[10]

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Q6.b CO-3; SO-2; BL-3		Two independent random samples of series are drawn from normal population with the same variance. The sample results are given below: $n_1 = 4, \bar{x} = 12, s_1 = 4$ $n_2 = 5, \bar{y} = 10, s_2 = 3$ where s_1^2 and s_2^2 are unbiased estimate of the common population variance. Also identify 90% confidence interval for $\mu 1 - \mu 2$	[10]
Q7.a CO-3; SO-2; BL-4		Based on field experiments, a new variety green gram is expected to give an yield of 12.0quintals per hectare. The variety was tested on 10 randomly selected farmers' fields. The yield (quintals/hectare) were recorded as 14.3,12.6,13.7,10.9,13.7,12.0,11.4,12.0,12.6,13.1. Do the results conform the expectation?	[10]
Q7.b CO-1; SO-2; BL-2		State Bayes theorem. Explain in detail with proof	[10]

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

5/5

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MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2022-2023

Program: B.Tech/MBATech Stream : Data Science — Year: II Semester: III

Subject: Managing Uncertainty — Time: 3 hrs (11:00am to 2:00pm)

Date: 14 / 06 / 2023 . — No. of Pages: 7

Marks: 100

Special Re-Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	20
CO- 1; SO-1 ; BL-1	A	The mean of the following frequency distribution was found to be 1.46 *No. of Accidents* — *No. of Days (frequency)* 0 — 46 1 — ? 2 — ? 3 — 25 4 — 10 5 — 5 Total — 200 days Calculate the missing frequencies.	5

CO-2 ; SO-2 ; BL-2	B	A brokerage survey reports that 30 percent of individual investors have used a discount broker, that is, one which does not charge the full commission. In a random sample of 9 individuals, what is the probability that I. Exactly two of the sampled individuals have used a discount broker II. Not more than three have used a discount broker III. At least three of them have used a discount broker	2 2 1
CO-4 ; SO-1 ; BL-2	C	There are five machines in a factory. Of these machines, three are working properly and two are defective. Machines which are working properly produce articles each of which has independently a probability of 0.1 of being imperfect. For the defective machines this probability is 0.2. A machine is chosen at random and five articles produced by the machine are examined	5
CO- 5; SO-2 ; BL-1	D	In a small business firm, two typists are employed-typist A and typist B. Typist A types out, on an average, 30 pages per day with a standard deviation of 6. Typist B, on an average, types out 45 pages with a standard deviation of 10. Which typist shows greater consistency in his output?	5
Q2		Answer briefly:	20
Q2 CO-1; SO-1; BL-1	A	Find the missing frequency in the following distribution, for which the median is 48.25 and mode, is 44.	10
Q2 CO-1; SO-1;		Calculate the most suitable average for the following data:	10

Class Interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90	Total
Frequency	3	7	?	?	?	20	16	4	150

Page 2

<table>
<tr><td>BL-2</td><td>B</td><td>Size of the Item Below 50 | 50-100 | 100-150 | 150-200 | 200 and above
Frequency 15 | 20 | 36 | 40 | 10</td><td></td></tr>
<tr><td>Q3</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q3
CO-2;
SO-2; BL-2</td><td>A</td><td>The following distribution relating to marks obtained by students in an examination:
Calculate Standard deviation of the marks
<table><tr><th>Marks</th><th>Number of Students</th></tr><tr><td>0- 10</td><td>1</td></tr><tr><td>10- 20</td><td>3</td></tr><tr><td>20- 30</td><td>6</td></tr><tr><td>30- 40</td><td>10</td></tr><tr><td>40- 50</td><td>12</td></tr><tr><td>50- 60</td><td>11</td></tr><tr><td>60- 70</td><td>6</td></tr><tr><td>70- 80</td><td>3</td></tr><tr><td>80- 90</td><td>2</td></tr><tr><td>90-100</td><td>1</td></tr></table></td><td>10</td></tr>
<tr><td>Q3
CO-2;
SO-1; BL-2</td><td>B</td><td>Calculate Karl pearson's coefficient of skewness.
<table><tr><th>Variable</th><th>Frequency</th><th>Variable</th><th>Frequency</th></tr><tr><td>70-80</td><td>11</td><td>30-40</td><td>21</td></tr><tr><td>60-70</td><td>22</td><td>20-30</td><td>11</td></tr><tr><td>50-60</td><td>30</td><td>10-20</td><td>6</td></tr><tr><td>40-50</td><td>35</td><td>0-10</td><td>5</td></tr></table></td><td>10</td></tr>
<tr><td>Q4</td><td></td><td>Answer briefly:</td><td>20</td></tr>
<tr><td>Q4
CO-;3
SO-2;
BL-1</td><td>A</td><td>Distribution of monthly income of a group of 3000 school teachers in a village conforms to a normal curve with $\mu = 600$ and $\sigma = 100$.
I. Find the percentage of teachers having monthly income of more than Rs. 800.
II. Number of teachers having monthly income less than Rs. 400.
III. Find the lowest monthly income amongst the highest paid 100 teachers.</td><td>10</td></tr>
</table>

		IV. The highest monthly income among the lowest paid teachers. V. Calculate the range of 3σ	
Q4 CO-;3 SO-2; BL-1	B	Consider a computer system with Poisson job-arrival stream at an average of 2 per minute. Determine the probability that in any one-minute interval there will be I. 0 jobs; II. exactly 2 jobs; III. at most 3 arrivals. IV. What is the maximum jobs that should arrive one minute with 90 % certainty? V. more than 3 Jobs	10
Q5		Answer briefly:	20
Q5 CO-4; SO-2; BL-2	A	For an experiment of throwing a die twice, find the probability: I. of the event of getting a total of 9, given that the die has shown up points between 4 and 6 (both inclusive) II. of the event of getting points between 4 and 6 (both inclusive), given that a total of 9 has already been obtained	5 5
Q5 CO-4; SO-2; BL-2	B	An Economist believes that during periods of high economic growth, the Indian Rupee appreciates with probability 0.70; in periods of moderate economic growth, it appreciates with probability 0.40; and during periods of low economic growth, the Rupee appreciates with probability 0.20. During any period of time the probability of high economic growth is 0.30; the probability of moderate economic growth is 0.50 and the probability of low economic growth is 0.20. Suppose the Rupee value has been appreciating during the present period. What is the probability that	10

		we are experiencing the period of (I) high, (II) moderate, and (III) low, economic growth?	
Q6		Answer briefly:	20
Q6 CO-5; SO-2; BL-3	A	The random variable X takes values -1, 0, 1 with probabilities 1/8, 2/8, 5/8 respectively. I. Compute E(X). II. Plot the PMF III. Give the pmf of $Y = X^2$ and use it to compute E(Y). IV. Instead, compute $E(X^2)$ directly from an extended table. V. Compute Var(X).	10
Q6 CO-5; SO-2; BL-2	B	Suppose now that events A, B and C are mutually independent with P(A) = 0.3, P(B) = 0.4, P(C) = 0.5. Compute the following: (Hint: Use a Venn diagram) I. $P(A \cap B \cap C^c)$ II. $P(A \cap B^c \cap C)$ III. $P(A^c \cap B \cap C)$	3 3 4
Q7		Answer briefly:	20

<table>
<tr>
<td>Q7
CO-2;
SO-2;
BL-3</td>
<td>A</td>
<td>The following table shows the results of two colleges. Which is better on the average?
<table>
<tr><td>Colleges →</td><td colspan="2">College A</td><td colspan="2">College B</td></tr>
<tr><td>Courses ↓</td><td>No.of students</td><td>Marks %</td><td>No.of students</td><td>Marks %</td></tr>
<tr><td>I degree</td><td>200</td><td>70</td><td>150</td><td>80</td></tr>
<tr><td>II degree</td><td>150</td><td>60</td><td>100</td><td>60</td></tr>
<tr><td>III degree</td><td>100</td><td>80</td><td>50</td><td>80</td></tr>
</table>
</td>
<td>10</td>
</tr>
<tr>
<td>Q7
CO-3; SO-2;
BL-3</td>
<td>B</td>
<td>Explain Type I and Type II error</td>
<td>10</td>
</tr>
</table>

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

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Programme: B. Tech (DS)/B. Tech Intg. (DS)/MBA Tech (DS)
Year: II / IV/Semester III/Semester VII(Exam Year: 2025-2026)
Subject: Managing Uncertainty
Date: 29 Nov 2025 **Time: 10:00 am to 01:00 pm (03:00 Hrs.)**
Max. Marks: 100

FINAL EXAMINATION (Acad. Year:**2025-2026**)

Instructions:

1. This question paper contains 4 pages.
2. Question No. 1 is compulsory.
3. Out of remaining questions, attempt any 4 questions.
4. In all 5 questions to be attempted.
5. All questions carry equal marks.
6. Answer to each new question to be started on a fresh page.
7. Figure in brackets on the right-hand side indicates full marks
8. Assume Suitable data if necessary.

<table>
<tr><td>Q1</td><td>Q.1</td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ; BL-3</td><td>a.</td><td>The following table gives the frequency distribution of the weekly wages (in '00 Rs.) of 100 workers in a factory.
<table>
<tr><td>Weekly Wages (₹00)</td><td>20–24</td><td>25–29</td><td>30–34</td><td>35–39</td><td>40–44</td><td>45–49</td><td>50–54</td><td>55–59</td><td>60–64</td></tr>
<tr><td>Number of Workers</td><td>4</td><td>5</td><td>12</td><td>23</td><td>31</td><td>10</td><td>8</td><td>5</td><td>2</td></tr>
</table>
Develop the histogram and frequency polygon of the distribution.</td><td>[5]</td></tr>
<tr><td>CO-1,5 ; BL-2</td><td>b.</td><td>Compare Binomial and Poisson Distribution</td><td>[5]</td></tr>
<tr><td>CO-2 ; BL-3</td><td>c.</td><td>A bag contains 7 white and 9 black balls. Two balls are drawn in succession without replacement at random. Identify the probability that one of them is white and the other is black</td><td>[5]</td></tr>
<tr><td>CO-1 ; BL-2</td><td>d.</td><td>Explain the Qualitative and Quantitative types of classification in detail</td><td>[5]</td></tr>
</table>

Q2.a CO-3,4 ; BL-4		Distinguish Karl Pearson's coefficient of skewness from the following data and explain its significance Wages (₹): 70–80 | 80–90 | 90–100 | 100–110 | 110–120 | 120–130 | 130–140 | 140–150 No. of Persons: 12 | 18 | 35 | 42 | 50 | 45 | 20 | 8	[10]
Q2.b CO-2 ; BL-4		Simplify and State Bayes' theorem and provide a detailed proof along with an explanation of its components and applications.	[10]
Q3 CO-3 ; BL-5		In the following data, two class frequencies are missing. Class Interval: 100-110 | 110-120 | 120-130 | 130-140 | 140-150 | 150-160 | 160-170 | 170-180 | 180-190 | 190-200 Frequency: 4 | 7 | 15 | f1 | 40 | f2 | 16 | 10 | 6 | 3 However, it was possible to ascertain that the total number of frequencies was 150 and that the median has been correctly found out as 146·25. You are required to Evaluate with the help of information given : (i) The two missing frequencies. (ii) Having found the missing frequencies, calculate arithmetic mean and standard deviation. (iii) Find the value of mode.	[20]
Q4.a CO-2,6 ; BL-5		The joint probability function of X and Y is given by $p(x,y)=k(x^2+y)$ for x=1,2 and y=1,2,3. Determine: (i) The value of k. (ii) The marginal distributions of X and Y. (iii) Are X and Y independent? (iv) The conditional distribution of X for Y=2.	[10]
Q4.b CO-2 ; BL-3		Ten unbiased coins are tossed simultaneously. Identify the probability of obtaining; (i) Exactly 6 heads (ii) At least 8 heads (iii) No head (iv) At least one head (v) Not more than three heads (vi) At least 4 heads.	[10]
Q5.a CO-2,5 ; BL-3		The time (in minutes), a customer spends waiting for a service agent at a call center follows an Exponential Distribution with a mean waiting time of 5 minutes. (a) Determine the probability density function (PDF) and the cumulative distribution function (CDF) for the waiting time, X. (b) Find the probability that a customer waits for more than 8 minutes. (c) Find the probability that a customer waits between 3 and 7 minutes.	[10]

<table>
<tr><td>Q5.b
CO-2 ;
BL-4</td><td></td><td>There are three identical-looking boxes: B1, B2, and B3. The composition of the balls in each box is as follows:
Box B1: 5 Red balls, 5 Green balls
Box B2: 2 Red balls, 8 Green balls
Box B3: 7 Red balls, 3 Green balls

A box is chosen at random, and then a ball is drawn from that box. The drawn ball is observed to be Red. Analyze is the probability that the ball came from Box B3?</td><td>[10]</td></tr>
<tr><td>Q6.a
CO-3,6 ;
BL-2</td><td></td><td>Explain Type I and Type II error. Explain Skewness and Kurtosis in detail</td><td>[10]</td></tr>
<tr><td>Q6.b
CO-3 ;
BL-3</td><td></td><td>The number of critical errors detected per hour on a new e-commerce website was observed over a 200-hour testing period.

<table><tr><td>Number of Errors per Hour (X)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>Observed Frequency (f)</td><td>120</td><td>55</td><td>18</td><td>5</td><td>2</td></tr></table>
Identify the expected frequencies using the Poisson distribution. Also, compute the mean and variance of the fitted (theoretical) distribution.</td><td>[10]</td></tr>
<tr><td>Q7.a
CO-2;
BL-5</td><td></td><td>A continuous random variable X has the following probability density function
$$f(x) = \frac{a}{x^5}, 2 \leq X \leq 10$$
Determine the constant a, distribution function of X and find the probability of the event $4 \leq X \leq 7$</td><td>[10]</td></tr>
<tr><td>Q7.b
CO-5,6 ;
BL-6</td><td></td><td>A manufacturer claims that the mean lifespan of their new LED bulb is 8,000 hours. A consumer testing agency randomly selected 12 bulbs and recorded their actual lifespans (in hundreds of hours):

78,85,75,82,79,88,80,81,74,86,83,76

Test whether these results support the manufacturer's claim? Use a significance level of α=0.01.</td><td>[10]</td></tr>
</table>

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										